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Numerical Methods to Simulate and Study Primordial Black Holes

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Chapter 1

Abstract

Using a mixed methods approach, this study aims to verify the results of Escrivà (2020) surrounding their Primordial Black Hole (PBH) simulations through numerical methods and cosmological perturbation theory. We investigate the behaviour of the system governed by the Misner-Sharp equations following a perturbation near criticality. Moreover, we achieve a deeper insight into the findings of this paper and verify their numerical results. We combine numerical techniques including bisection methods, spectral methods with Chebyshev Polynomials and the 4th order Runge-Kutta technique for assisting time evolution. For decades, the threshold defining when a perturbation in the early Universe would lead to a collapse into a PBH has been studied and we find an agreeing value to literature of $\delta_c \approx 0.49774 \pm 2 \times 10^{-5}$. Applying the spectral methods and Runge-Kutta results in qualitative analysis, through visual inspection of the time evolution and sonification of particular values to further our understanding of collapse or dispersion. We establish that collapsing behaviour is not steady, but rather a rapidly accelerating process, while the converse is true for dispersion, a decelerating process. Initiated by the sonification results, we establish a fitted function describing the collapse as a function of time in the supercritical regime. This research shall ultimately be applied to deepening our knowledge of both dark matter and the early Universe within the first second of the Big Bang.

Chapter 2

Introduction

2.1 Historical Background and Motivation

Primordial Black Holes (PBHs) were first hypothesised 60 years ago by Zel'dovich & Novikov (1966) as bodies in the early stages of the expansion of the Universe. Unlike astrophysical black holes, which arise from stellar collapse, PBHs theoretically encode information about the early Universe shortly after the Big Bang and have since been at the forefront of research in cosmology and general relativity (GR). Early formulations suggested a potentially catastrophic conflict with standard cosmological evolution, however, this issue was later resolved by abandoning the assumption of static accretion. We now know that if PBHs were formed, they would have lived in an expanding Friedmann-Robertson-Walker (FRW) universe, rather than in isolation. Motivated by these corrections provided by Carr & Hawking (1974), physicists have been dedicated to finding a robust model of PBHs and their evolution, with the first PBH formation within the FRW universe being correctly adapted by Shibata & Sasaki (1999). While theoretically, PBHs obey the mathematical laws of GR and fluid mechanics, their highly non-linear nature severely limits purely analytic approaches. As a result, an efficient method for studying their behaviour and evolution lies in simulations and approximations.

In order to simulate and study a PBH numerically, we must understand where and when such a black hole could form. The relevant parameter is called the critical threshold, δ_c. This threshold is a measure of a perturbation generated by the early Universe during cosmic inflation. More specifically, this is the minimum amplitude of a super-horizon curvature perturbation such that, upon horizon re-entry, gravity exactly overcomes pressure and expansion, causing black hole formation instead of dispersion. Namely, below the threshold the perturbation disperses and above, a black hole is formed. At the threshold, the evolution approaches a self-similar critical solution. While the threshold value depends weakly on the shape of the initial curvature profile, the behaviour itself is universal due to this self-similarity at the critical value (Kehagias et al., 2025). Mathematically, the critical threshold is a value derived from the *compaction function*, a function which depends on a radial term and mass terms. The relevant mass terms are FRW background mass and Misner-Sharp mass, a quasi-local measure of gravitational energy in a spherically symmetric spacetime. The role of Misner-Sharp mass becomes clear as we introduce the governing relativistic fluid mechanical equations, known as the Misner-Sharp equations (Misner & Sharp, 1964). These equations will later allow us to study the evolution of the

system where the fluid describes the early Universe.

To quantify the critical threshold, one must first specify the background cosmology and the dynamics applied in evolving a perturbation through horizon re-entry. This is when a perturbation that was initially larger than the cosmological horizon becomes smaller than it, as the horizon grows with time. We primarily study PBH formation under the assumption of spherical symmetry in order to capture the collapse in its dominant terms, while also allowing for the simulation to be computationally efficient. Another assumption that is made within the FRW background, is that the Universe is filled by a perfect fluid characterised by a fixed equation-of-state parameter. The Misner-Sharp formalism, therefore, provides a natural mathematical basis for the evolving system. This system is rather complex as we are joining a gravitational system with a hydrodynamical system. Following a perturbation in the initial stages of the system, cosmological perturbation theory serves as a mathematical tool to describe the propagation of such an effect. Understanding when the perturbation leads to collapse involves deriving a physical link to the system. In particular, we look at characterising velocity, density and the effect the perturbation has on curvature. The study of PBHs is increasingly relevant to modern research as they present as a promising candidate for dark matter. Due to their small size (approximately of the atomic scale), they can contain the mass of a large mountain (in the order of 10^{12} kg (Frampton, 2016)). However, theoretically some could be several solar masses. Due to their expected evaporation by Hawking radiation, they could take several billion years to evaporate, and thus could still exist around the Universe today. Therefore, by understanding their formation, and thus their life and evolution, we gain insight into both the early Universe and seeking evidence for dark matter (Green, 2024).

Motivated by the recent work by Escrivà (2020), we combine the approaches to re-derive and extend the numerical methods to analyse the critical threshold to better understand the physics of PBH formation and what this means for the Universe.

The project is structured as follows, Section 2.2 will outline the mathematical background which will be the basis for later calculations, Section 3.1 shall detail the cosmological adaptation of the described mathematics and will introduce cosmological perturbation theory as a means of quantifying the collapse or dispersion threshold. This is followed by Section 3.2 which explains the numerical methods applied and how the simulation is performed. We continue with Section 4.1 and Section 4.2 covering results and discussions, including a brief note on error and uncertainty. Finally we shall conclude in Section 5.

2.2 Mathematical Background

Ultimately, the aim is to translate a mathematical model of the system to a numerical simulation. We structure the mathematical picture as follows, understanding the GR within the system, expanding a perturbation within the system with cosmological perturbation theory, using a bisection method to identify the critical threshold and finally spectral methods are used to simplify and replace spatial derivatives in the Misner-Sharp equations to aid the numerical time evolution of the perturbation.

2.2.1 Spherical Symmetry and Misner Sharp Equations

Considering an ideal fluid with the following energy momentum tensor given by,

$$T^{\mu\nu} = (p+\rho)u^\mu u^\nu + pg^{\mu\nu}, \tag{2.1}$$

where u^μ is the four-velocity, p is the pressure, ρ is the energy-density and $g^{\mu\nu}$ is the metric, we describe the energy flow and pressure in all directions (Schutz, 2009). One can relate p and ρ with an equation-of-state scaled by the equation-of-state parameter. The corresponding line element to the metric is given by,

$$\mathrm{d}s^2 = -A(r,t)^2\,\mathrm{d}t^2 + B(r,t)^2\,\mathrm{d}r^2 + R(r,t)^2\,\mathrm{d}\Omega_2, \tag{2.2}$$

where $\mathrm{d}\Omega_2 = \mathrm{d}\theta^2 + \sin^2(\theta)\,\mathrm{d}\phi^2$, the line element of a 2-sphere and $R(r,t)$ is the areal radius (as opposed to coordinate radius) which relates position with the expansion of the Universe and is a geometric invariant. The remaining functions, $A(r,t)$ and $B(r,t)$ are to be determined from a later expansion. Einstein's perturbed equations of motion are given by,

$$R_{\mu\nu} - \frac{1}{2}g_{\mu\nu}R = 8\pi(T_{\mu\nu} + \Delta T_{\mu\nu}), \tag{2.3}$$

where $\Delta T_{\mu\nu}$ denotes an initial perturbative variation of the energy momentum tensor, and the local energy and momentum conservation is given by,

$$\nabla_\nu(T^{\mu\nu} + \Delta T^{\mu\nu}) \equiv (T^{\mu\nu} + \Delta T^{\mu\nu})_{;\nu} = 0,$$

(Misner & Sharp, 1969). Since we established that the stress-energy tensor depends on density and pressure, it is clear that the effective perturbation would be an effect on this fluid. Following the methods outlined in Misner & Sharp (1964), we introduce the variables

$$U(r,t) \equiv D_t R = \frac{1}{A(r,t)}\frac{\partial R(r,t)}{\partial t}, \tag{2.4}$$

and

$$\Gamma(r,t) \equiv D_r R = \frac{1}{B(r,t)}\frac{\partial R(r,t)}{\partial r}, \tag{2.5}$$

where D_t and D_r denote the co-moving proper time and proper distance derivatives, respectively. The quantity U measures the radial motion of the fluid relative to the coordinate origin, and is therefore the radial component of the four-velocity in the Eulerian frame (Escrivà, 2020). $\Gamma(r,t)$ is a measure of the radial change in space-time as a result of the curvature, but it can loosely be thought of as a local curvature measure.

Returning to the previously mentioned Misner-Sharp mass, we define it as,

$$M(r,t) \equiv \int_0^R 4\pi R^2 \rho \left(\frac{\partial R}{\partial r}\right) \mathrm{d}r. \tag{2.6}$$

Thus, the full set of field equations, the three first order dynamical equations, along with the remaining

derivatives, collectively the Misner-Sharp equations, are given by,

$$D_t R = U, \tag{2.7}$$

$$D_t M = -4\pi R^2 p U, \tag{2.8}$$

$$D_t U = -\left[\frac{\Gamma}{(p+\rho)} D_r p + \frac{M}{R^2} + 4\pi R p\right], \tag{2.9}$$

$$D_t \rho = -\frac{\rho + p}{\Gamma R^2} D_r (U R^2), \tag{2.10}$$

$$D_r M = 4\pi \Gamma \rho R^2, \tag{2.11}$$

$$D_r A = -\frac{A}{\rho + p} D_r p. \tag{2.12}$$

The set of equations describes the evolution of a collapse with spherical symmetry in non-linear full GR. We also apply the boundary conditions enforcing $R(0,t) = U(0,t) = M(0,t) = 0$. Due to the spherical symmetry, $D_r p(0,t) = 0$ (Escrivà, 2020). The initial conditions are not arbitrary, but rather result from a controlled asymptotic expansion. We are concerned with studying the effect of a perturbation on this set of equations which will be rigorously derived and detailed in a later section. We also wish to constrain where apparent horizon is formed. In particular, this appears when considering the gradient of R becoming null. This is written as,

$$g^{\mu\nu} \nabla_\mu R \nabla_\nu R = 0,$$

which is a measure of the total energy contained in a sphere of radius R (Misner & Sharp, 1964). The left hand side can also be written explicitly as

$$1 - \frac{2M}{R} = g^{\mu\nu} \nabla_\mu R \nabla_\nu R, \tag{2.13}$$

where the Misner-Sharp mass is now geometrically defined. The apparent horizon formation condition is therefore

$$2M/R = 1. \tag{2.14}$$

This corresponds to the location where the outgoing null geodesics have zero expansion, marking the boundary of the trapped region (Carr & Hawking, 1974).

The set of equations along with the additional constraints and conditions form the foundation for the PDEs which have to be evolved. Later we will discuss the spectral methods applied to the GR based FRW background.

2.2.2 Critical Threshold

Cosmologically, the mass defined with Equation (2.13) is divided between a background FRW mass (M_b) and a contribution due to the perturbation manipulating the mass to become a dense region. This motivates the definition of the *compaction function* given by,

$$C(r) = 2\frac{M(r) - M_\mathrm{b}(r)}{R(r)}. \tag{2.15}$$

The compaction function is a gauge-invariant which quantifies the local mass excess per unit radius (Musco et al., 2005), and may be interpreted as a local density. As the perturbation evolves and re-enters the horizon, the compaction function reaches a maximum. The point at which this maximum is obtained, r_m, identifies the region prone to collapse, namely the effective size of the space which will become a PBH. We will later consider r_m to be the expansion scale and its relevance to any approximations we make. When $r = r_m$, the mass excess (or gravitational pull) is strongest, relative to its size (Hawking & Ellis, 1973). Similarly, the pressure gradients are optimally balanced. To relate the compaction function to the critical threshold in the *long-wavelength regime* (which will be formally defined in the following section), we assign $C(r_m) = \bar{\delta}(r_m)$ in order to define the critical threshold as

$$\bar{\delta}(r_m) \geq \delta_c, \tag{2.16}$$

which follows from what was first discovered by Choptuik (1993). The apparent horizon forms when the expansion of outgoing null geodesics vanishes, which in spherical symmetry is equivalent to the condition given in Equation (2.14).

Chapter 3

Cosmological Applications and Adjustments

3.1 Background Cosmology and Approximations

Another essential to mathematically quantify is a consistent initialisation of a perturbation within an FRW universe which will satisfy Einstein's equations (2.3) (Weinberg, 2008). As introduced above, the appropriate regime for PBHs far outside of the cosmological horizon is called the long-wavelength expansion, a controlled asymptotic expansion of the Einstein-fluid equations in which spatial gradients are treated as small compared to time derivatives, valid when the physical size of a perturbation is much larger than the cosmological horizon. We initially wish to consider the FRW background with energy-density assumed to be sourced by a perfect fluid with the fixed equation-of-state parameter. This parameter appears in the equation-of-state $p = \omega\rho$ as ω. From this setting, the cosmological horizon can define a scale from which causal interactions are suppressed. We therefore consider perturbations with perturbation scales initially larger than the horizon (Shibata & Sasaki, 1999).

3.1.1 Cosmological Context

Regarding the context of cosmology, following the formal set up by Einstein, we turn to the work of Stephen Hawking, and in particular later research with Bernard Carr (Carr & Hawking, 1974). Before we can establish approximations and dynamics within this cosmological picture, we must first understand how we can make a somewhat agnostic metric and set of equations describe this cosmological situation in particular. In modern cosmology, the early Universe is understood as an epoch in which radiation was the dominant component, as opposed to matter or cosmological constant. In particular, Carr & Hawking (1974) conclude that the modern Universe implies that the early Universe must be somewhat inhomogeneous. This relates to density fluctuations, which could decouple from cosmic inflation and gravitationally collapse. Later we will describe the mathematical picture in which this fits, called the FRW metric, though it is important to note that any approximations and methods we select must follow this description of the early Universe. More precisely, the assumptions surrounding homogeneity which defy initial theories are precisely the contradiction which will reconcile PBHs and their formation.

3.1.2 Long-Wavelength and Misner-Sharp Equations

To investigate the non-linear evolution and formation of PBHs from a collapse in this particular cosmological regime, the fluid equations combined with the implications from Einstein's equations must be reduced with cosmological perturbation theory. Motivated by the work of Escrivà (2020), we aim to ultimately determine the critical threshold of the system by a bisection method. The pseudo-spectral technique applies Chebyshev spectral methods and 4th order Runge-Kutta to numerically evolve the system following a perturbation from the critical threshold, or near it. We must express the parameters related to the PBH in terms of the critical threshold to study its behaviour. This adapted version of the Misner-Sharp equations is the set of equations within the context of the FRW background. The use of the above equation of state closes the Misner-Sharp system and ensures consistency with an initially homogeneous FRW background. As the radial variable approaches infinity, we want this to reflect the FRW background, but for a numerical approach, we must constrain the system within finite intervals. Following the logic detailed in Escrivà (2020), we use the condition $D_r p(r_f, t) = 0$ where the cut-off r_f is the maximum for our finite representation of an infinite radius. Analytically we can solve for $A(r,t)$, in the neighbourhood of $r = r_f$. Equation (2.12) yields,

$$\int \frac{D_r A}{A} = -\int \frac{D_r p}{\rho + p},$$

where the equation-of-state with constant ω implies,

$$\ln A = -\int \frac{\omega}{\omega + 1} \frac{D_r \rho}{\rho}.$$

Integrating, we obtain,

$$A(r,t) = \left(\frac{\rho_b(t)}{\rho(r,t)} \right)^{\frac{\omega}{\omega+1}},$$

where we have fixed the constant of integration, the energy-density of the FRW background, as $\rho_b(t) = \rho_0 (t_0/t)^2$ with $\rho_0 = 3H_0^2/8\pi$, where H_0 is the Hubble expansion rate at time t_0. Rewriting the proper time and distance derivatives (Equations (2.7)-(2.12)) as standard time and radial derivatives for a subsequent Taylor expansion, we implement Equations (2.4) and (2.5) as,

$$D_t \chi = \frac{1}{A(r,t)} \dot{\chi},$$

and

$$D_r \chi = B(r,t) \chi',$$

where χ is any one of the variables described. We are particularly interested in the time derivatives for the later expansion, but the radial derivatives would follow in a similar way. Starting with rewriting Equation (2.9), dropping the argument of the functions for clarity, we find,

$$\dot{U} = -A \left[\frac{\Gamma}{(p+\rho)} D_r p + \frac{M}{R^2} + 4\pi R p \right], \tag{3.1}$$

Since we are treating ω as constant, $D_r P = \omega D_r \rho$, which can be replaced by $B\rho'$. As we have no analytic form for B, we write,

$$B\frac{\partial \rho}{\partial r} = B\frac{\partial \rho}{\partial R}\frac{\partial R}{\partial r} = B\frac{\partial \rho}{\partial R}R'.$$

Recall Equation (2.5) defines Γ. Thus,

$$D_r \rho = \Gamma \frac{\partial \rho}{\partial R} = \Gamma \frac{\rho'}{R'}.$$

Combining the above result with our systems equation-of-state, we find Equation (3.1) becomes,

$$\dot{U} = -A\left[\frac{\omega \Gamma^2 \rho'}{(1+\omega)\rho R'} + \frac{M}{R^2} + 4\pi R \omega \rho\right]. \tag{3.2}$$

The remaining quantities follow in a similar and less involved way, given by

$$\dot{R} = AU, \tag{3.3}$$

$$\frac{\dot{B}}{B} = A\frac{U'}{R'}, \tag{3.4}$$

$$\dot{\rho} = -A\rho(1+\omega)\left(\frac{2U}{R} + \frac{U'}{R'}\right), \tag{3.5}$$

$$\dot{M} = -4\pi A \omega \rho U R^2 = -4\pi \omega \rho R^2 \dot{R}, \tag{3.6}$$

$$\frac{A'}{A} = -\frac{\omega}{1+\omega}\frac{\rho'}{\rho}, \tag{3.7}$$

$$\frac{R'^2}{B^2} = 1 + U^2 - \frac{2M}{R} = \Gamma^2. \tag{3.8}$$

where ($\dot{}$) refers to time derivatives and ($'$) refers to spatial derivatives. Equation (3.8) is directly what we call the constraint equation, from which the variables U, Γ, and M,

$$\Gamma = \sqrt{1 + U^2 - \frac{2M}{R}}. \tag{3.9}$$

Notice that when $\Gamma = |U|$, the horizon constraint emerges. This Γ constraint can directly be derived from Equation (2.13). This follows $\partial_\mu R = (\dot{R}, R')$, with,

$$g^{tt} = -\frac{1}{A^2}, \qquad g^{rr} = \frac{1}{B^2}.$$

Therefore,

$$1 - \frac{2M}{R} = -\frac{1}{A^2}\dot{R}^2 + \frac{1}{B^2}R'^2,$$

which by Equations (2.5) and (3.2), becomes exactly Equation (3.9). Also notice that if $M = 0$ (flat space-time), Γ is a parallel to the Lorentz factor in special relativity.

The spatial derivatives shall be addressed with spectral methods and the time derivatives shall be treated with fourth-order Runge Kutta techniques to study their evolution with time. For an initial perturbation in the associated fluid, however, we shall implement cosmological perturbation theory.

3.1.3 Cosmological Perturbations and Expansions

Consider the metric in Equation (2.2) as super-horizon scales, namely the FRW metric, at leading orders in gradient expansion,

$$\mathrm{d}s^2 = -\,\mathrm{d}t^2 + s^2(t)\left[\frac{\mathrm{d}r^2}{1-K(r)r^2} + r^2\,\mathrm{d}\Omega_2\right], \tag{3.10}$$

where $s(t)$ is the scale factor and $K(r)$ is a curvature profile. At leading order, $K(r)r^2$ is proportional to the compaction function, Equation (2.15). As we are in the long-wavelength approximation, we write our expansion parameter, ϵ, with some dimensional analysis to instil time dependence, as $\epsilon(t) \equiv (R_H(t)/s(t)r_m)^2$, ($\epsilon^2$ in some texts). We expand the relevant variables at leading order results in a perturbative expression with $\epsilon(t) << 1$, precisely the long-wavelength approximation. Comparing the general functions A and B from the previous metric to this one and Taylor expanding the main quantities applying the derivatives found above returns the following variables:

$$A(r,t) = 1 + \epsilon(t)\tilde{A}(r), \tag{3.11}$$

$$B(r,t) = \frac{R'(r,t)}{\sqrt{1-K(r)r^2}}\left(1+\epsilon(t)\tilde{B}(r)\right), \tag{3.12}$$

$$R(r,t) = s(t)r\left(1+\epsilon(t)\tilde{R}(r)\right), \tag{3.13}$$

$$U(r,t) = H(t)R(r,t)\left(1+\epsilon(t)\tilde{U}(r)\right), \tag{3.14}$$

$$\rho(r,t) = \rho_b(t)\left(1+\epsilon(t)\tilde{\rho}(r)\right), \tag{3.15}$$

$$M(r,t) = \frac{4\pi}{3}\rho_b(t)R(r,t)^3\left(1+\epsilon(t)\tilde{M}(r)\right). \tag{3.16}$$

($\tilde{}$) variables shall be computed next. Note that as ϵ approaches 0, the regular FRW solution variables are restored. While A and B are not relevant to our study of the evolution and behaviour of PBHs, they are helper functions in finding the perturbative expansions of the remaining quantities.

Bridging the formulation outlined in Polnarev & Musco (2007), except we shall instead maintain a constant for ω, we first must establish the correlation with the scale factor in the super-horizon scale. Since $\rho_0 = 3H_0^2/8\pi$ describes the energy-density when $K(r) = 0$, and the relation between the Hubble parameter and the scale factor is

$$H(t) = \frac{\dot{s}}{s}, \tag{3.17}$$

we have

$$\left(\frac{\dot{s}}{s}\right)^2 = \frac{8\pi}{3}\rho_0.$$

Taking the covariant derivative of the energy momentum tensor in Equation (2.1) applying the covariant conservation law, $\nabla_\mu T^{\mu\nu} = 0$, and considering the homogeneous/isotropic perfect fluid is co-moving in the coordinates where $u^\mu = (1,0,0,0)$ and $u_\mu = (-1,0,0,0)$, we find,

$$\dot{\rho} + 3H(\rho + p) = 0.$$

Using the equation-of-state and the Hubble parameter relation,

$$\frac{\dot{\rho}_b}{\rho_b} = -3(1+\omega)\frac{\dot{s}}{s}. \tag{3.18}$$

Following our definition of the expansion parameter ϵ, we can write,

$$\frac{\dot{\epsilon}}{\epsilon} = -2\frac{\dot{s}}{s} - \frac{\dot{\rho}_b}{\rho_b} = (1+3\omega)\frac{\dot{s}}{s}. \tag{3.19}$$

Returning to the ($\tilde{}$) variables in Equations (3.11)-(3.16), we shall define an independent time variable $\zeta = \ln s$. For the quantities, we can find a time derivative by,

$$\dot{\tilde{\chi}} = \frac{\dot{s}}{s}\frac{\partial}{\partial\zeta}\tilde{\chi}. \tag{3.20}$$

Starting with Equation (3.3), and applying the expansions for each of the variables to first order (using Equations (3.13), (3.11) and (3.14)), we must take the time derivative of Equation (3.13). The left hand side of the expansion follows,

$$\dot{R} = \dot{s}r(1+\epsilon\tilde{R}) + sr(\dot{\epsilon}\tilde{R} + \dot{\tilde{R}}\epsilon), \tag{3.21}$$

$$\frac{\dot{R}}{sr} = \frac{\dot{s}}{s}(1+\epsilon\tilde{R}) + (\dot{\epsilon}\tilde{R} + \dot{\tilde{R}}\epsilon).$$

Expanding the right hand side of Equation (3.3) gives,

$$AU = (1+\epsilon\tilde{A})HR(1+\epsilon\tilde{U}) = (1+\epsilon\tilde{A})Hsr(1+\epsilon\tilde{R})(1+\epsilon\tilde{U}),$$

where H is replaced by Equation (3.17), thus,

$$AU = \dot{s}r(1+\epsilon(\tilde{A}+\tilde{R}+\tilde{U})).$$

Dividing both sides by s and equating to the left hand side from before, upon cancelling terms we find,

$$\dot{\tilde{R}} + \frac{\dot{\epsilon}}{\epsilon}\tilde{R} = \frac{\dot{s}}{s}(\tilde{A}+\tilde{U}).$$

Combining the results from Equations (3.19) and (3.20), we have,

$$\frac{\partial}{\partial\zeta}\tilde{R} + (1+3\omega)\tilde{R} = \tilde{A}+\tilde{U}. \tag{3.22}$$

Applying a similar logic to the remaining variables, we aim to represent them all as simple systems of partial differential equations (PDEs). As an example for the solving of a resulting PDE, we follow the perturbative expansion of Equation (3.6). Staring with taking the time derivative of Equation (3.16),

$$\dot{M} = \frac{4\pi}{3}\left\{\dot{\rho}_b R^3(1+\epsilon\tilde{M}) + 3\rho_b R^2\dot{R}(1+\epsilon\tilde{M} + \rho_b R^3(\epsilon\tilde{M})^{\cdot}\right\}.$$

Factorising and comparing to the unperturbed right hand side, we find some appear on both sides,

$$\frac{4\pi}{3}\rho_b R^3 \left\{ \frac{\dot{\rho}_b}{\rho_b}(1+\epsilon\tilde{M}) + 3\frac{\dot{R}}{R}(1+\epsilon\tilde{M}) + (\epsilon\tilde{M})^{\cdot} \right\} = -4\pi\omega\rho R^2\dot{R}$$

Using Equation (3.15) to apply the perturbative expansion on the right, and implementing Equation (3.18), to first order in ϵ, we obtain,

$$\frac{1}{3}(1+\epsilon\tilde{M})\left(-3(1+\omega)\frac{\dot{s}}{s} + 3\frac{\dot{R}}{R} + (\epsilon\tilde{M})^{\cdot}\right) = -\omega(1+\epsilon\tilde{\rho})\frac{\dot{R}}{R}.$$

Expanding both sides, we find,

$$-3(1+\omega)\frac{\dot{s}}{s} + 3(1+\omega)\frac{\dot{R}}{R} - 3(1+\omega)\frac{\dot{s}}{s}\epsilon\tilde{M} + 3\frac{\dot{R}}{R}\epsilon\tilde{M} + 3\frac{\dot{R}}{R}\epsilon\tilde{\rho} + (\epsilon\tilde{M})^{\cdot} = 0,$$

precisely what is given in Polnarev & Musco (2007). Replacing the ratio with $\dot{R}$ and R with their respective expansions in Equations (3.21) and (3.13), linearising in ϵ, we can replace the ratio with,

$$\frac{\dot{s}}{s} + \dot{\epsilon}\tilde{R} + \dot{\tilde{\epsilon}}R,$$

thus, to first order in ϵ,

$$\begin{aligned} 3(1+\omega)\left(\frac{\dot{R}}{R} - \frac{\dot{s}}{s}\right) + (\epsilon\tilde{M})^{\cdot} + 3\epsilon\tilde{M}\left(\left(\frac{\dot{R}}{R} - \frac{\dot{s}}{s}\right) - \omega\frac{\dot{s}}{s}\right) + 3\omega\frac{\dot{R}}{R}\epsilon\tilde{\rho} \\ \simeq \dot{\epsilon}\tilde{M} + \epsilon\dot{\tilde{M}} + 3(1+\omega)(\dot{\epsilon}\tilde{R} + \epsilon\dot{\tilde{R}}) + 3\omega\frac{\dot{s}}{s}\epsilon(\tilde{\rho} - \tilde{M}) = 0. \end{aligned}$$

By virtue of replacing any variable with both a tilde and a time derivative with the result of Equation (3.20) and a collection of terms becoming exactly Equation (3.22), we arrive at,

$$\frac{\partial\tilde{M}}{\partial\zeta} + \tilde{M} + 3((1+\omega)\tilde{A} + \omega\tilde{\rho}) = -3(1+\omega)\tilde{U}. \tag{3.23}$$

One of the derivative equations that we have yet to expand is Equation (3.7). Expanding to linear order and integrating to find,

$$(1+\omega)\tilde{A} + \omega\tilde{\rho} = 0,$$

precisely the bracketed term in Equation (3.23). Therefore, we have simplified the PDE to,

$$\frac{\partial\tilde{M}}{\partial\zeta} + \tilde{M} = -3(1+\omega)\tilde{U}. \tag{3.24}$$

The remaining variables are perturbed in a similar way, though the result we shall implement to further simplify and solve the PDE in Equation (3.24) is the result from expanding Equation (3.8). With the expansion terms involving the curvature $K(r)$ from Equation (3.12), with the logic and

explanation from Polnarev & Musco (2007), we arrive at,

$$-K(r)r^2 = \frac{R^2\dot{s}^2}{s^2}(2\tilde{U} - \tilde{M}). \tag{3.25}$$

Comparing this with the constraint equation (Equation (3.9)) allows us to write the curvature in terms of Γ,

$$K(r) = \frac{1-\Gamma^2}{r^2}.$$

Rearranging Equation (3.25) to isolate $\tilde{U}$, writing this in Equation (3.24), and rewriting $\epsilon = 1/(\dot{s}^2 r_m^2)$, since we can relate the scale factor to Hubble parameter and therefore directly to the cosmological horizon. As a result, we obtain the PDE in terms of curvature instead,

$$\frac{\partial \tilde{M}}{\partial \zeta} + \frac{5+3\omega}{2}\tilde{M} = \frac{3}{2}(1+\omega)K(r)r_m^2. \tag{3.26}$$

Since ζ is effectively a time variable, and curvature does not depend on time (and the rest of the quantities are constants), we can separate the variables as,

$$\tilde{M} = f(\zeta)K(r)r_m^2, \tag{3.27}$$

where we have now to solve an ordinary differential equation (ODE) for $f(\zeta)$ instead. This looks like,

$$\frac{\mathrm{d}f}{\mathrm{d}\zeta} + \frac{5+3\omega}{2}f = \frac{3}{2}(1+\omega).$$

In a similar way, we can write $\tilde{\rho}$, $\tilde{A}$ and $\tilde{U}$ in terms of $f(\zeta)$ as well (Polnarev & Musco, 2007),

$$\tilde{\rho} = f(\zeta)\frac{1}{3r^2}(r^3K(r))'r_m^2, \tag{3.28}$$

$$\tilde{A} = -f(\zeta)\frac{\omega}{1+\omega}\frac{1}{3r^2}(r^3K(r))'r_m^2, \tag{3.29}$$

and

$$\tilde{U} = \frac{1}{2}(f(\zeta)-1)K(r)r_m^2. \tag{3.30}$$

Diverging from the steps laid out in Polnarev & Musco (2007), we shall continue with the assumption that ω remains constant. We fix the constants of integration using results from cosmological perturbation theory. We will show that for this constant, our expressions for $f(\zeta)$ are consistent.

Solving $f(\zeta)$ analytically with the integration factor, we are solving,

$$f'(\zeta) + Pf(\zeta) = Q,$$

where $P = \frac{5+3\omega}{2}$, and $Q = \frac{3}{2}(1+\omega)$. The integrating factor, $\exp(P\zeta)$, means we must integrate,

$$\left(e^{P\zeta}f\right)' = Qe^{P\zeta}.$$

Rearranging for $f(\zeta)$, gives,

$$f(\zeta) = \frac{3(1+\omega)}{(5+3\omega)} + Ce^{-\frac{5+3\omega}{2}\zeta}.$$

Recall that ζ is related to t as $\zeta = \ln(s)$. Therefore $\exp(-P\zeta) = \exp(-P\ln(s)) = s^{-P}$. When t approaches 0, the scale factor, s, also approaches zero. Therefore if C is non-zero, and ω is greater than zero, Cs^{-P} blows up. For a radiation dominated Universe, namely the time-frame for which PBHs would form, ω is larger than 0 (more precisely, as will be relevant for the simulations, $\omega = 1/3$ for a radiation dominated Universe). Thus,

$$f(\omega) = \frac{3(1+\omega)}{5+3\omega}, \tag{3.31}$$

is constant in our regime[1]. Since the PDE involved with finding $\tilde{R}$ in Equation (3.22) depends on $\tilde{A}$ and $\tilde{U}$, in terms of f, this becomes,

$$\frac{\partial}{\partial\zeta}\tilde{R} + (1+3\omega)\tilde{R} = -f\frac{\omega}{1+\omega}\frac{1}{3r^2}(r^3K(r))'r_m^2 + \frac{1}{2}(f-1)K(r)r_m^2.$$

We shall use a similar trick to above, introducing $g(\zeta)$ and $h(\zeta)$. This allows us to write the above equation as,

$$\tilde{R} = -g(\zeta)\frac{1}{3r^2}(r^3K(r))'r_m^2 + h(\zeta)\frac{K(r)}{r}r_m^2, \tag{3.32}$$

where we must instead solve the following ODEs by integrating factor,

$$\frac{\mathrm{d}g}{\mathrm{d}\zeta} + (1+3\omega)g = \frac{\omega}{1+\omega}f(\omega),$$

$$\frac{\mathrm{d}h}{\mathrm{d}\zeta} + (1+3\omega)h = f(\omega) - 1.$$

Repeating the integrating factor and the logic for setting the integration constant to zero, inputting the constant form for $f(\omega)$, we find that in the radiation dominated regime, indeed we drop the ζ dependence altogether and have found the omega dependent factors,

$$g(\omega) = \frac{3\omega}{(1+3\omega)(5+3\omega),} \tag{3.33}$$

$$h(\omega) = -\frac{2}{(1+3\omega)(5+3\omega)}. \tag{3.34}$$

With Equations (3.31), (3.33) and (3.34), we find Equation (3.27), Equations (3.28)–(3.30) and Equa-

[1] In a radiation dominated Universe, a well established ω value is 1/3, thus $f(\omega) = 2/3$.

tion (3.32), return to those which Escrivà (2020) summarises as,

$$\tilde{\rho}(r) = \frac{3(1+\omega)}{5+3\omega}\left(K(r) + \frac{r}{3}K'(r)\right) r_m^2, \tag{3.35}$$

$$\tilde{U}(r) = -\frac{1}{5+3\omega}K(r)r_m^2, \tag{3.36}$$

$$\tilde{A}(r) = -\frac{\omega}{1+\omega}\tilde{\rho}(r), \tag{3.37}$$

$$\tilde{M}(r) = -3(1+\omega)\tilde{U}(r), \tag{3.38}$$

$$\tilde{R}(r) = -\frac{\omega}{(1+3\omega)(1+\omega)}\tilde{\rho}(r) + \frac{1}{1+3\omega}\tilde{U}(r). \tag{3.39}$$

when we reinstate their radial dependence.

3.1.4 Averaged Density Contrast

Following the notation convention of Escrivà (2020), starting with the Misner-Sharp mass in Equation (2.6), we define the averaged density contrast inside an areal radius $R(r,t)$ as

$$\delta(r,t) = \frac{1}{V}\int_0^R 4\pi R^2 R'(r,t)\frac{\delta\rho}{\rho_b}\,\mathrm{d}r = \frac{1}{V}\int_0^R 4\pi R^2 \frac{\rho-\rho_b}{\rho_b}R'\,\mathrm{d}r, \qquad V = \frac{4\pi R^3}{3}.$$

Using $\rho = \rho_b(1+\epsilon\tilde{\rho})$, this becomes

$$\delta(r,t) = \frac{3}{R^3(r,t)}\int_0^r \epsilon(t)\,\tilde{\rho}(\bar{r})\,R^2(\bar{r},t)R'(\bar{r},t)\,\mathrm{d}\bar{r}.$$

In the long-wavelength limit, $R(r,t) \simeq a(t)r$, so to leading order

$$\delta(r,t) \simeq \epsilon(t)\frac{3}{r^3}\int_0^r \tilde{\rho}(\bar{r})\,\bar{r}^2\,\mathrm{d}\bar{r}.$$

Using the quasi-homogeneous solution

$$\tilde{\rho}(r) = \frac{3(1+\omega)}{5+3\omega}\left(K(r) + \frac{r}{3}K'(r)\right) r_m^2,$$

and the identity

$$\left(K + \frac{r}{3}K'\right) r^2 = \frac{1}{3}\frac{d}{dr}(r^3 K),$$

the integral reduces to

$$\delta(r,t) = \epsilon(t)\frac{3(1+\omega)}{5+3\omega}K(r)\,r_m^2.$$

Identifying the long-wavelength parameter $\epsilon(t) = (aHr_m)^{-2}$, we obtain

$$\delta(r,t) = \left(\frac{1}{aHr_m}\right)^2 \bar{\delta}(r), \qquad \bar{\delta}(r) = f(\omega)K(r)\,r_m^2,$$

which matches the result of Escrivà (2020). This implies $C(r) = f(\omega)K(r)r^2 = r^2\bar{\delta}(r)/r_m^2$. Thus evaluating around $r = r_m$ unites this conclusion with that of Equation (2.16). Thus, once we find

the critical threshold, $\bar{\delta}(r_m)$ numerically, we can study the behaviour of a perturbation leading to dispersion or a collapse into a PBH.

Since r_m is specifically the value for which $C(r)$ is maximum, we can study $C'(r_m) = 0$. This implies,

$$f(\omega)K'(r_m)r_m^2 + 2f(\omega)K(r_m)r_m = 0,$$

or explicitly, since $f(\omega) \neq 0$, the equation given by,

$$K'(r_m) + \frac{2}{r_m}K(r_m) = 0, \tag{3.40}$$

around the stationary-point, $r = r_m$. We can fit an exponential curvature profile with this condition, namely,

$$K(r) = K_0 e^{-\frac{1}{q}(r/r_m)^{2q}}, \tag{3.41}$$

where $q = 1$ reduces to a Gaussian curvature, satisfying $K(0) = K_0$. Despite having some freedom for the curvature profile, we select from those relevant in cosmology (Germani & Musco, 2019). Around the stationary point, Equation (3.41) satisfies Equation (3.40). By a bisection method to be discussed in the following section, if a maximum for $C(r_m)$ is found, we solve for the corresponding curvature scaling K_0 value,

$$K_0 = \frac{\delta_c}{f(\omega)K(r_m)r_m^2},$$

which allows for the full system to be studied under an initial perturbation. Once the value of K_0 is found, corresponding to the critical threshold, all profiles can be studied as we have established that all of the variables depend on the curvature.

3.2 Numerical Methods

3.2.1 Bisection for Critical Threshold

To find δ_c numerically, we implement a bisection method up to a given tolerance. Defining a reasonable lower and upper bound for the critical threshold around literature values ($\delta_c \approx 0.5$ (Musco, 2019)), we pick $\delta_{c,\min} = 2/5$ and $\delta_{c,\max} = 2/3$ to begin, and can adjust to accelerate the bisection process. Aligned with the standard of the bisection method, we shall start with the midpoint value, evaluate the compaction function at this value and evaluates the maximum. If $C_{\max} \geq 1$, by Equation (2.14), a horizon has already been formed and thus a black hole has already formed, so we must check beneath this. We take $C_{\min} = 0.3$ (Escrivà, 2020) as this is a subcritical value and thus disperses the perturbation. We fine-tune each iterative δ_c by the midpoints shifting with new values of δ_c approaching a collapse, up to a certain tolerance. We shall see that a critical value is reached when the compaction function's amplitude remains constant over time evaluated at different radial positions. We are restricted to a resolution (difference between iterative input and output δ_c) of order $\mathcal{O}(10^{-5})$, due to the development of large pressure gradients upon approach to the critical, self-similar solution. Namely, ρ'/R' blows up when the resolution becomes too small, causing failure within the numerics.

3.2.2 Pseudo-Spectral Techniques

We adopt a pseudo-spectral numerical method which combines Chebyshev polynomials and Lagrange polynomials. Using basis functions to simplify a problem into a numerical one, while adding the leverage of alternative analytical results is a common technique, in particular in astronomy and cosmology (Martinez et al., 2025). For the problems we encounter with describing evolution of time within a specific system, basis functions are an advantage to study a somewhat analytic function, rather than just numerical values. Chebyshev polynomials provide a natural basis for functions defined on finite intervals and avoid the introduction of artificial periodic boundary conditions. Spectral methods are particularly well suited to this problem, as the solutions are smooth and the dynamics are dominated by global behaviour rather than sharp local features. Alternative and perhaps more trivial methods to approach the same problem include a technique called 'null slicing', which studies the outgoing or ingoing null hyper-surfaces as opposed to space-like ones. This technique naturally accommodates for the horizon formation but is less suited to a situation with a cosmological system where long-wavelength conditions (and therefore the expanding FRW background) is required (Shibata & Sasaki, 1999). We are specifically interested in studying the time evolution of the spatial and time derivative equations (Equations (3.2)-(3.8)) following the initial perturbation.

In order to derive the functions applied to the spectral methods, we first consider an arbitrary function $f(x)$ approximated by N Chebyshev polynomials. Mathematically, this is written as,

$$f_N(x) = \sum_{k=0}^{N} a_k T_k(x), \tag{3.42}$$

where a_k are fit coefficients to be found and each $T_k(x)$ is a Chebyshev polynomial of the first kind of degree k,

$$T_k(\cos\theta) = \cos(k\theta), \quad \text{where } x = \cos\theta.$$

The coefficients are solved by introducing collocation points at,

$$x_n = \cos\left(\frac{n\pi}{N}\right),$$

the Chebyshev-Lobatto collocation nodes, and solving $f_N(x_n) = f(x_n)$, where the approximated function is equal to the exact function. To do this, we define the *nodal polynomial* which vanishes at the collocation points as,

$$\ell(x) = \prod_{j=0}^{N}(x - x_j).$$

For Chebyshev-Lobatto nodes, note that,

$$(1 - x_n^2)T_N'(x_n) = 0, \quad 1 \leq n \leq N - 1,$$

since $T_N'(x_n) = 0$ as well as the zeroes in the prefactor at $x_n = \pm 1$. Hence,

$$\ell(x) = c(1 - x^2)T_N'(x),$$

for some constant c. Introducing the Lagrange polynomials defined in terms of $\ell(x)$, the nth Lagrange function is given by,

$$L_n(x) = \frac{\ell(x)}{(x - x_n)\ell'(x_n)}, \tag{3.43}$$

where,

$$\ell'(x) = c[-2xT_N'(x) + (1 - x^2)T_N''(x)].$$

Since $T_N'(x_n) = 0$ at the nodes,

$$\ell'(x_n) = c(1 - x_n^2)T_N''(x_n). \tag{3.44}$$

Computing $T_N''(x)$ by employing the Chebyshev polynomials of the second kind and considering the result when $\theta = \theta_n = n\pi/N$, where $\sin(N\theta_n) = 0$ and $\cos(N\theta_n) = (-1)^n$, returns,

$$T_N''(x_n) = \frac{N^2(-1)^n}{1 - x_n^2}.$$

Thus, at the collocation nodes, Equation (3.44) becomes,

$$\ell'(x_n) = -cN^2(-1)^n\bar{c}_n, \quad \bar{c}_n = \begin{cases} 2, & n = 0, N, \\ 1, & \text{otherwise.} \end{cases}$$

As a result, Equation (3.43) is given by,

$$L_n(x) = \frac{(-1)^{n+1}\,(1 - x^2)\,T_N'(x)}{\bar{c}_n\,N^2\,(x - x_n)}, \qquad \bar{c}_n = \begin{cases} 2, & n = 0, N, \\ 1, & \text{otherwise.} \end{cases} \tag{3.45}$$

The solution to Equation (3.42) is,

$$f_N(x) = \sum_{n=0}^{N} L_n(x)f(x_n),$$

whose k^{th} derivative is simply,

$$f_N^{(k)}(x_i) = \sum_{n=0}^{N} L_n^{(k)}(x_i)f_N(x_n). \tag{3.46}$$

As we are interested in numerically solving the equations governing the motion of the fluid, the spatial derivatives present a challenge in solving the PDE system. Given the nodal values $f(x_n)$ at the Chebyshev-Lobatto points, there exists a unique interpolating polynomial $f_N(x)$ of degree at most N. The differentiation matrix then gives the derivative of the interpolating polynomial evaluated at the collocation points. Rather than approximating the solution only pointwise, we approximate it by a global function that captures its behaviour across the whole domain. Appendix A outlines the mathematical background to implement the differentiation matrix $\mathbf{D}$ (we drop the superscript denoting which derivative it is as we are only interested in the first derivative, we also use ($'$) to represent this first derivative). This appendix also details the translation into python code. Because $\mathbf{D}$ acts on

closed-form expressions for Chebyshev-Lobatto nodal values, and $L_k(x_j) = \delta_{jk}$, the derivative of the Lagrange polynomial in Equation (3.46) can be replaced in the following way,

$$L'_k(x_i) = \sum_j D_{ij} L_k(x_j). \tag{3.47}$$

In particular,

$$D_{ik} = \frac{L'_k(x_i)}{L_k(x_k)}.$$

Since the denominator is 1 due to the Kronecker-delta condition, we replace each $L(x_i)$ component with a matrix element of $\mathbf{D}$. Thus, we obtain the discrete relation,

$$f'_N(x_i) = \sum_{j=0}^{N} D_{ij}\, f(x_j).$$

Specifically, for a function we can write,

$$f' = \mathbf{D} f.$$

We can therefore write explicitly any partial spatial derivative component-wise, for example,

$$R'_i = \sum_j D_{ij} R_j,$$

can be substituted into the relevant equations. Thus, our system of time derivative PDEs which depend on spatial derivatives (Equations (3.2) and (3.5)), are now discretised differential equations, which for our simulation can be treated as Ordinary Differential Equations (ODEs), given by,

$$\dot{U}_i = -A_i \left[\frac{\omega \Gamma_i^2 \sum_j D_{ij}\rho_j}{(1+\omega)\rho_i \sum_j D_{ij} R_j} + \frac{M_i}{R_i^2} + 4\pi R_i \omega \rho_i \right],$$

$$\dot{\rho}_i = -A_i \rho_i (1+\omega) \left(\frac{2U_i}{R_i} + \frac{\sum_j D_{ij} U_j}{\sum_j D_{ij} R_j} \right),$$

where Equations (3.3) and (3.6) remain the same as they only depend on time derivatives. Again we are omitting the quantity B in the analysis as it does not provide further insight than what we already have with the other variables. The upper limit on the sum, shall be set to N, the order of polynomials to sum over.

The above definitions are translated into Python and their time evolution is solved with fourth order Runge-Kutta (Escrivà, 2020).

3.2.3 Runge-Kutta Technique

Once the spatial derivatives have been discretised, the Misner-Sharp equations reduce to a system of ODEs in time of the form

$$\frac{dX}{dt} = F(X, t),$$

where X represents the set of dynamical variables (U, R, M, ρ) evaluated at the collocation points. To evolve this system in time we employ the classical fourth-order Runge-Kutta method.

For a general equation $dy/dt = f(y, t)$, the Runge-Kutta scheme advances the solution from t_n to $t_{n+1} = t_n + \Delta t$ by computing four intermediate slopes

$$\begin{aligned}
k_1 &= f(y_n, t_n), \\
k_2 &= f\left(y_n + \frac{\Delta t}{2} k_1, t_n + \frac{\Delta t}{2}\right), \\
k_3 &= f\left(y_n + \frac{\Delta t}{2} k_2, t_n + \frac{\Delta t}{2}\right), \\
k_4 &= f(y_n + \Delta t\, k_3, t_n + \Delta t),
\end{aligned}$$

and updating the solution according to

$$y_{n+1} = y_n + \frac{\Delta t}{6}\left(k_1 + 2k_2 + 2k_3 + k_4\right).$$

This method provides fourth-order accuracy in the time-step Δt and is widely used in numerical relativity due to its stability and efficiency for evolving non-linear dynamical systems (Press et al., 2007). We input a particular value of Δt to the code and tune the maximum number of iterations to complete. Therefore, with the time derivatives found with the spectral methods, we can evolve over small intervals, meaning we now have all the tools for implementing the simulation and studying the behaviour of PBHs.

Chapter 4

Results and Discussion

The main physical results are products of the numerics and simulations. We will follow by studying the numerical value achieved for δ_c and comparing its effects on the dynamics of the system. While most of the Python code is sourced from the source Escrivà (2020) paper on GitHub[1], some changes have been made to the structure, notation and definitions. All code for plots is original as this was not given in the source GitHub. Our original notebook, with the plotting software, as well as animations and any sonification is available in the following public access GitHub repository https://github.com/meganhmartinez/PBHCapstone.

4.1 Numerical Computations

Our objectives in terms of numerical computations are three-fold, achieving a value for δ_c, reproducing similar plots to those in Escrivà (2020) and physically interpreting the graphs we find to study the dynamics of such a PBH.

4.1.1 Bisection to find the Critical Threshold

Implementing the upper and lower limits and performing the bisection code directly from Escrivà (2020) confirms their claim that $\delta_c \approx 0.49774 \pm 2 \times 10^{-5}$ is achieved. Appendix B specifically details the output and approach to this particular value. With the given initial conditions, this took over 13 minutes to run. Considering we find literature values around $\delta_c \approx 0.5$ (Musco, 2019), we can significantly simplify the conditions to find a faster approach in lower order of precision, to agree with literature. Though, we aim to characterise this physical value to more precision. While we may not change the resolution much, the process can be accelerated by starting with a smaller range of maximum and minimum threshold values. By changing the tolerance to 10^{-2} and reducing the maximum Chebyshev order by half, we find $\delta_c \approx 0.483 \pm 0.016$, a value which remains close to literature computed in under two minutes. A simulated value to higher precision which agrees with literature claims adds credibility and emphasises the suitability of this particular approach, allowing for the progression of the simulation.

[1] https://github.com/albert-escriva/SPriBHoS

4.1.2 Simulations

As we have established the critical threshold, we can study three distinct cases of perturbations. We shall label the inputted perturbation as $\bar{\delta}_m$. The first, subcritical ($\bar{\delta}_m = 0.49$ in our simulation), a perturbation which shall disperse and not cause a black hole to form. The second case is the critical case, with an initial perturbation related to the critical threshold we found in the previous section, balancing between collapse and dispersion. And lastly a super-critical case (we take $\bar{\delta}_m = 0.51$), when the perturbation undergoes collapse. While the Python code which generates the plots in Escrivà (2020) is not included in their repository, we were able to regenerate their graphs at particular time intervals from the quantities in the source code. Similar to the corresponding figures in Escrivà (2020), we shall study the relation between the following quantities and position in space at different points in time: the compaction function, the density function, the velocity 'U' measuring the radial velocity of the fluid, and 'Γ', the measurement of how radial distances stretch and compress within the curvature. Following the initial perturbation, as described in Section 3.2.2 and Section 3.2.3, we must apply spectral methods and Runge-Kutta techniques to evolve the functions with time, and study their behaviour in the three cases. For the Runge-Kutta computations, to maintain comparable results to the source, Escrivà (2020), we shall use a conformal time-step defined around the expansion of the Universe. In an expanding Universe, the scale factor $s(t)$, sometimes called $a(t)$ in cosmology, behaves like t^α, where in a matter-dominated Universe, $\alpha = 2/3$ and in a radiation dominated Universe, $\alpha = 1/2$. Thus our time step should behave like this also. Each time step, $\mathrm{d}t$, is written as,

$$\mathrm{d}t = \mathrm{d}t_0 \left(\frac{t}{t_0}\right)^\alpha,$$

where we will maintain dimensionless units, setting $t_0 = 1$. The remaining initial conditions are defined in the following way, in parallel to Escrivà (2020). Setting both t_0 and $s(t_0) = s_0$ equal to one, we find $H_0 = 1/2$ and $R_H(t_0 = 1) = 2$. In order to satisfy the long wavelength approximation, we have set $r_m = 10R_H(t_0)$, resulting in $t_m = 10^2 t_0$ (Polnarev & Musco, 2007). While Escrivà (2020) maintain a maximum Chebyshev limit of $N = 400$, we were able to improve computation time while simultaneously maintaining comparable results in the plots. The full set of initial conditions are compiled in Appendix C.

Analysing whether a perturbation undergoes dispersion or collapse, the compaction function evaluated at different time intervals provides the first clear indicator. Figure 4.1 shows the subcritical, critical and supercritical case at the same time intervals. Starting with the subcritical case, a decrease in the amplitude is visible with time. This decrease in amplitude is the visualisation of the dispersion of the perturbation. The critical case displays a constant amplitude, with the peak at the critical value. The roughly constant amplitude of the function across the time snapshots is related to the system sitting at the threshold between dispersion and collapse, which of course is visible in the rapidly increasing amplitude in the compaction function. Such behaviour is characteristic of gravitational collapse, indicating that the perturbation grows sufficiently over-dense to continue contracting toward black hole formation. Also, notice that the radial position of the peak is quickly moving to the left, namely where $r = 0$, thus indicating localisation of the collapsing region toward the origin.

Furthermore, we shall continue our study with the three cases by comparing the density function.

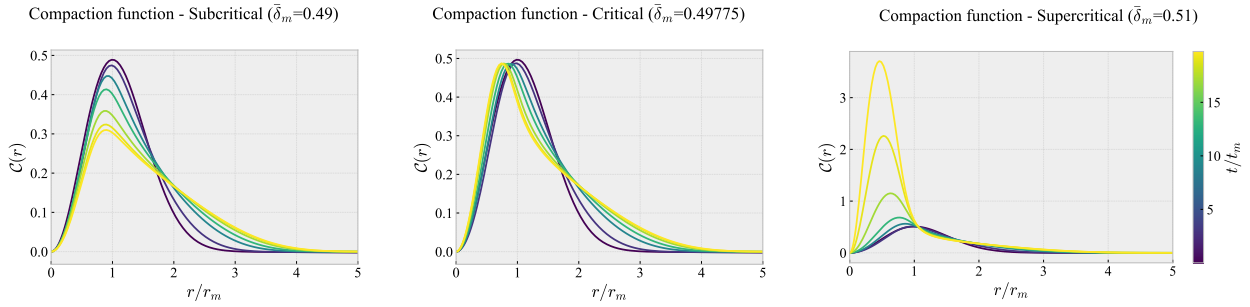

Figure 4.1: Compaction function for subcritical (left), critical (middle) and supercritical (right) cases.

Figure 4.2 shows the density profile (with a log scale) relative to the cosmological background, where ρ_b, the FRW background density. Where $\rho/\rho_b > 1$, we find an 'over density'. In all three cases, the central density increases with time, indicating that matter is initially drawn inward toward the over-dense region, thus increasing the density further. However, the rate and extent of this growth differ significantly between the three regimes. In the subcritical case, the increase in the central density is relatively modest and the profile remains broad, indicating that the perturbation does not become sufficiently concentrated to undergo collapse. Also, to further analyse the dispersion, after reaching a maximum density following the perturbation, the profile begins to decrease again, visible in the last two time snapshots overlapping, denoting both the growth and subsequent decay phases. As time progresses, pressure gradients in the radiation fluid drive an outward flow which redistributes the mass. This causes the central density to decrease and eventually fall below the background value, producing an under-dense region near the origin. At the same time, the displaced matter forms a broad over-dense shell near the perturbation scale r_m. This evolution further solidifies that the perturbation does not collapse but instead disperses, with the density profile gradually relaxing toward the background FRW solution at large radii. Figure 4.3 presents this behaviour in the longer time limit.

In the critical case, the peak grows more strongly and becomes narrower, reflecting behaviour close to the threshold separating dispersion from gravitational collapse. As the mass concentrates due to the perturbation, but not enough to constitute collapse, the density contrast grows dramatically near the origin as the perturbation evolves. However, instead of forming an apparent horizon, the system approaches a near-critical configuration where gravitational attraction and pressure gradients remain finely balanced. The outer region remains close to the background value $\rho/\rho_b \approx 1$, demonstrating that the dynamics are confined to a small region around the perturbation scale. At large time scales, represented by Figure 4.3, a shallow under-dense region surrounding the central peak is also visible, reflecting the compensated nature of the perturbation as matter flows inward toward the collapsing core. In the supercritical case, the density contrast rises dramatically by several orders of magnitude at small radii, producing a sharply localised central peak. This strong central concentration is characteristic of runaway gravitational collapse and is consistent with the formation of a primordial black hole. In all panels, the outer region remains close to the background value $\rho/\rho_b \approx 1$, showing that the collapse dynamics are highly localised. All three cases (with subcritical showing the most pronounced behaviour) exhibit a dip below 1 at intermediate radii. This indicates the formation of an under-dense region surrounding the central over-density. As matter flows inward towards the centre of the

perturbation, the surrounding region temporarily loses mass, producing a rarefaction shell around the collapsing core. Such compensated profiles are typical of cosmological perturbations, where the growth of a central over-density is accompanied by a neighbouring under-density (Polnarev & Musco, 2007; Musco et al., 2009). This can be considered as the system's desire to return to its background equilibrium state. We consider the subcritical case to mirror a damped oscillator and expect this to be consistent across dynamical profiles.

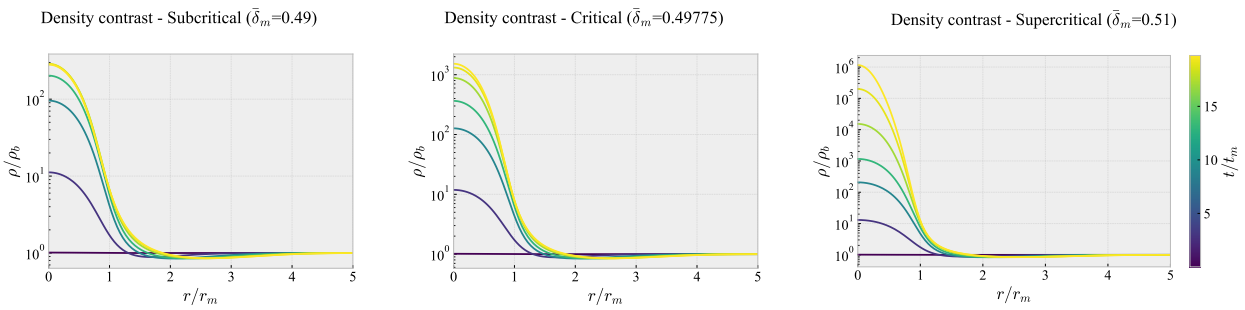

Figure 4.2: Density maps for subcritical (left), critical (middle) and supercritical (right) cases.

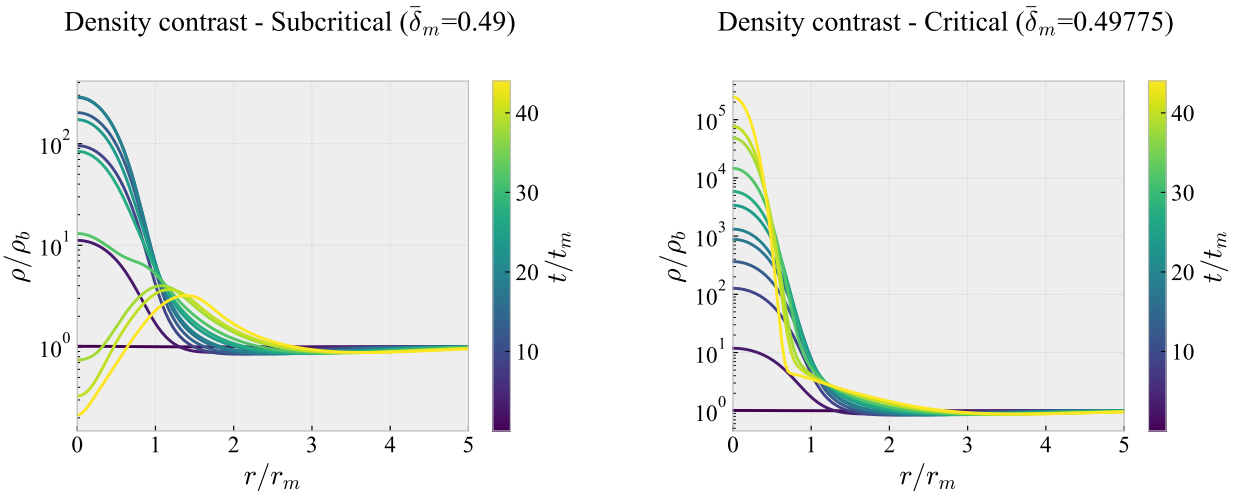

Figure 4.3: Long time visualisation of the density profile for subcritical (left) and critical (right) cases.

Likewise, examining the Γ profile gives insight to the behaviour resulting from the perturbation near the critical threshold. We interpret Γ as the stretching of space, while similar to curvature, it is better described as the strain and stress the spatial aspect of the system is under. Deviations from unity indicate deviations from FRW background. Figure 4.4 displays the three cases starting at 1 when r approaches 0 at all times, this follows the limit resulting from Equation (3.9). Another common behaviour across the three plots is a minimum on or near the perturbation scale r_m. Regardless of the case, the existence of a perturbation requires the largest mass excess to be at r_m. Namely, the gravitational mass excess is largest and the spacetime curvature is strongest. In the subcritical case, the dispersion leads to the dip becoming shallower with time. Figure 4.5 again exhibits damped oscillatory motion as the pressure pulls outwards above 1, at the same time scale that we see under-density, and

then the peak lowers again. In larger time scales, we expect the perturbation to fully disperse, and thus Γ would settle to 1, back in the FRW background. The critical case begins with a similar inwards pull, but instead of returning to the FRW background, the minimum stabilises, thus indicating a balance between gravity and pressure.

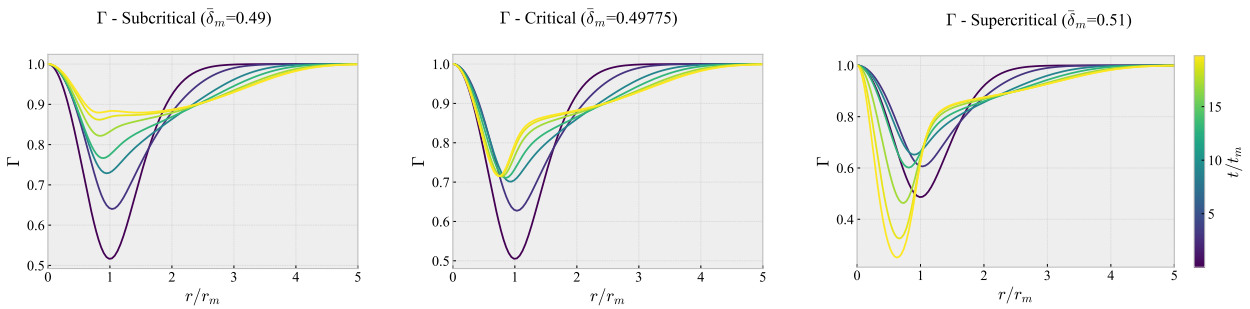

Figure 4.4: Γ function for subcritical (left), critical (middle) and supercritical (right) cases.

There is an inward distortion within the profile, indicating a split to inward and outward moving matter. In fact, if we study a longer time frame, we see the split more drastically in the inwards and outwards moving material, seen in Figure 4.5. Interestingly, the supercritical case in Figure 4.4 demonstrates a bounce following the initial perturbative pull on the space-time. As the perturbation compresses, the density and pressure increase rapidly near the centre, producing strong pressure gradients that temporarily drive an outward motion in the fluid. This generates a rarefaction wave which partially relaxes the spacetime curvature, causing a brief increase in Γ. However, since the perturbation is supercritical, gravitational attraction continues to dominate and indeed, the collapse proceeds toward black hole formation. It also agrees with our expectation of the development of strong curvature near the collapsing region. However, the fast moving curvature contradicts the assumptions under which the long-wavelength approximation is implemented (that gradients are small and that curvature evolves slowly), thus the approximation breaks down during the evolution (Escrivà, 2020). This also reflects the 'hole' developing and this is a singularity and thus numerically undefined.

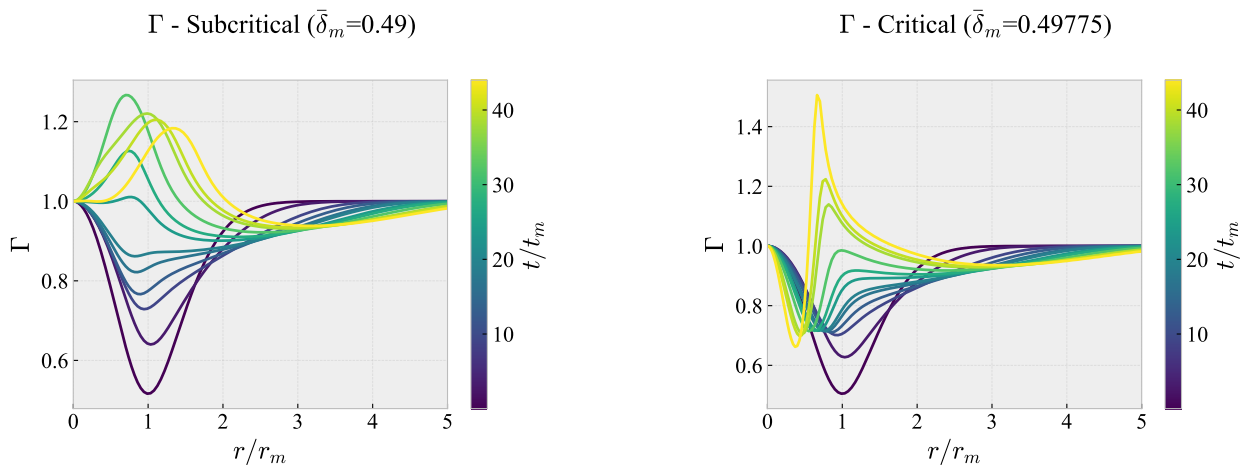

Figure 4.5: Long time visualisation of Γ profile for subcritical (left) and critical (right) cases.

With regards to the velocity profile, interesting behaviour is characterized by a change in direction. Dividing by Γ effectively normalises the velocity relative to the curvature. Figure 4.6 compares the three cases. The three cases share some similarities such rapid initial deceleration in the early stages and asymptotics to $U/\Gamma = 1$ at large times and distances, signature of the outer region being FRW, following cosmic expansion. The subtle differences in each case are indicative of the unique behaviour. In the subcritical case, negative values appear in the beginning, but after that, do not appear again. The critical case looks very similar to the subcritical, however the split into an inward and outward moving fluid is visible again, as later times have a positive and negative peak. The rarefaction and compression becomes more visible at larger times, however this behaviour is characteristic of the nature of the criticality. Finally, in the supercritical case where a PBH forms, the velocity becomes increasingly negative near the centre, indicating rapid inward motion associated with gravitational collapse, showing the infalling matter in that region.

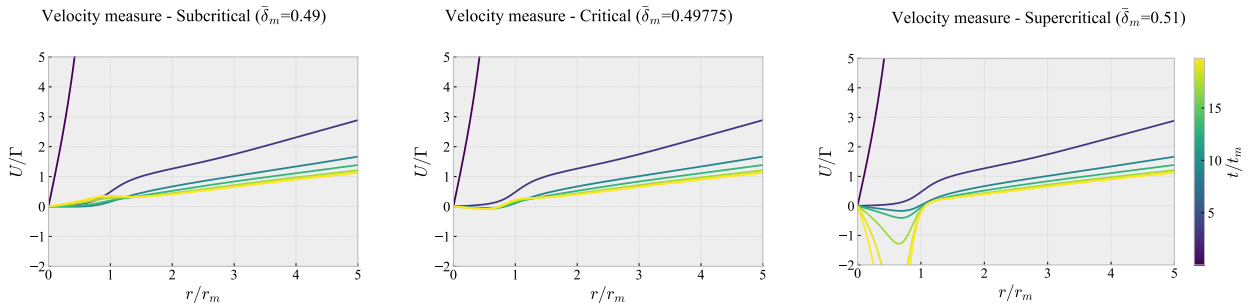

Figure 4.6: Velocity-Γ ratio for subcritical (left), critical (middle) and supercritical (right) cases.

Unfortunately, there is a maximum time limit on how far into the collapse process we can study the supercritical case in all four profiles, due to how fast the compression happens. Thus the code cannot simulate until the actual black hole formation, however it's the stages leading up to the formation which give insight to their nature. The analysis and graphs produced in this section matches exactly what Escrivà (2020) claims, adding credibility to their results and comments, despite studying different time intervals. We study the same time intervals for each case as a one-to-one comparison of behaviour across the threshold, as opposed to their isolated study of each case.

Upon observing the three cases, we gain deep understanding of the dynamics of the various profiles, even at small changes in $\bar{\delta}_m$. As a comparison, let us study extreme cases, an extreme subcritical taken for $\bar{\delta}_m = 0.2$, and an extreme supercritical around $\bar{\delta}_m = 0.65$. The four profiles for the extreme subcritical case, shown in Figure 4.7, reflect the perturbation quickly dispersing. The density quickly increases with the perturbation, but settles back to the FRW with an over-dense and under-dense front slowly returning to equilibrium. The compaction function exhibits the dispersion travelling radially over time. The Γ profile quickly returns to small oscillations around the equilibrium, with the velocity also approaching a linear constant speed. Similar to the subcritical case studied in detail above, similar, but more drastic, behaviour is seen, with the overall dynamics mirroring an over-damped oscillator.

In contrast, the extreme supercritical case, pictured in Figure 4.8, exhibits a substantially more rapid and pronounced collapse. The four profiles behave like the supercritical case studied previously, but in a much shorter time frame, and in a more drastic way. Interestingly, a steep peak manifests

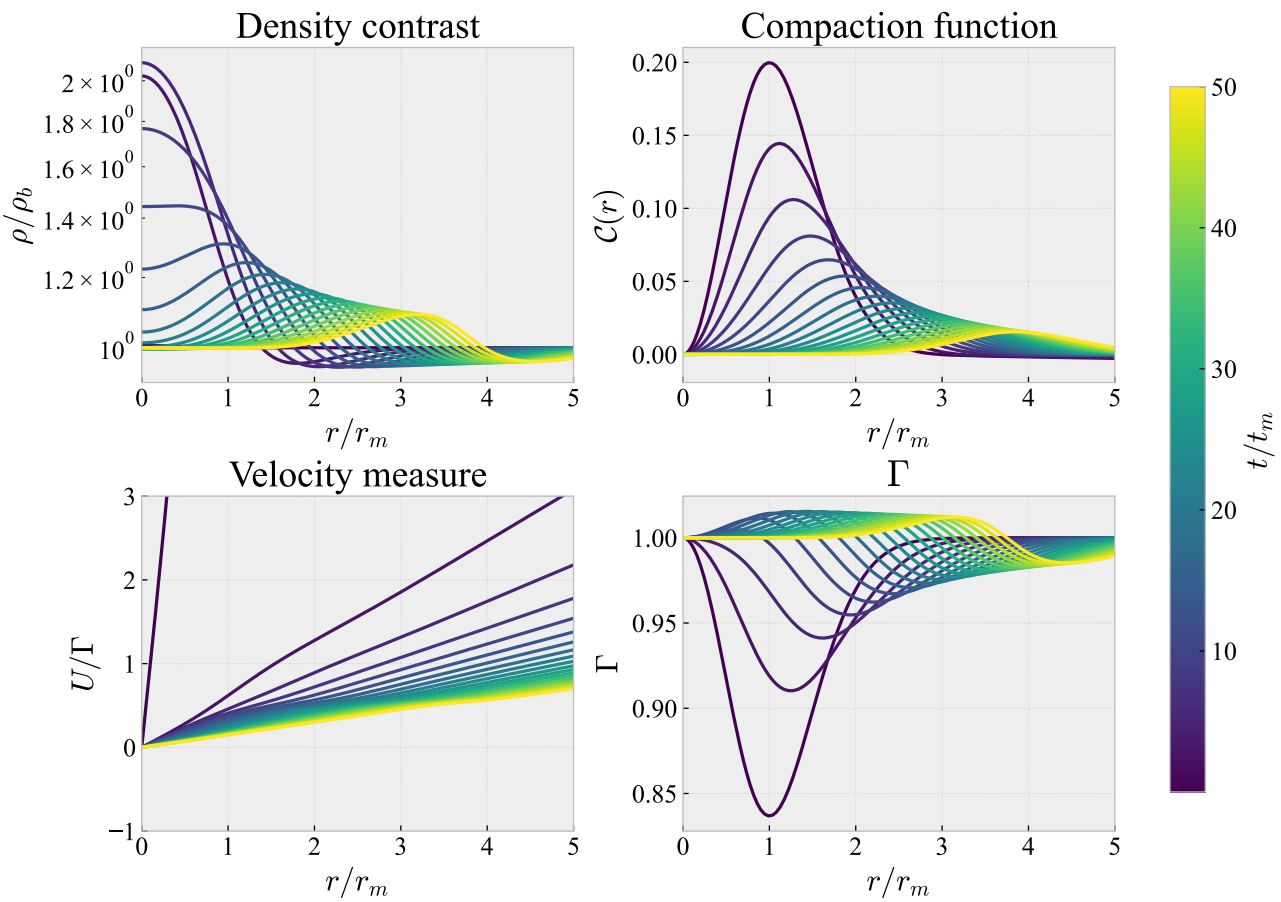

Figure 4.7: All profiles for extreme subcritical ($\bar{\delta}_m = 0.2$) case. Density (upper left), compaction function (upper right), velocity (lower left) and Γ (lower right).

within the velocity profile before the inwards movement. Similarly, the geometric factor Γ develops a deep minimum, reflecting the strong spacetime curvature produced by the growing mass concentration. Together, these features clearly signal runaway gravitational collapse leading toward black hole formation. Although these plots were not included in Escrivà (2020), they are useful to comprehend the detailed dynamics near and far from the criticality. Studying the stages leading to the collapse of a PBH gives us more intuition around this possible behaviour when the Universe could be described by such equations. The numerics and mathematics become a tool in this comprehension, rather than the point of study itself. Given that the analysis and results align with our main source, as well as other theoretical predictions (Polnarev & Musco, 2007; Musco et al., 2009), we have established a robust method of simulating this behaviour, while also providing a strong level of intuition around the results.

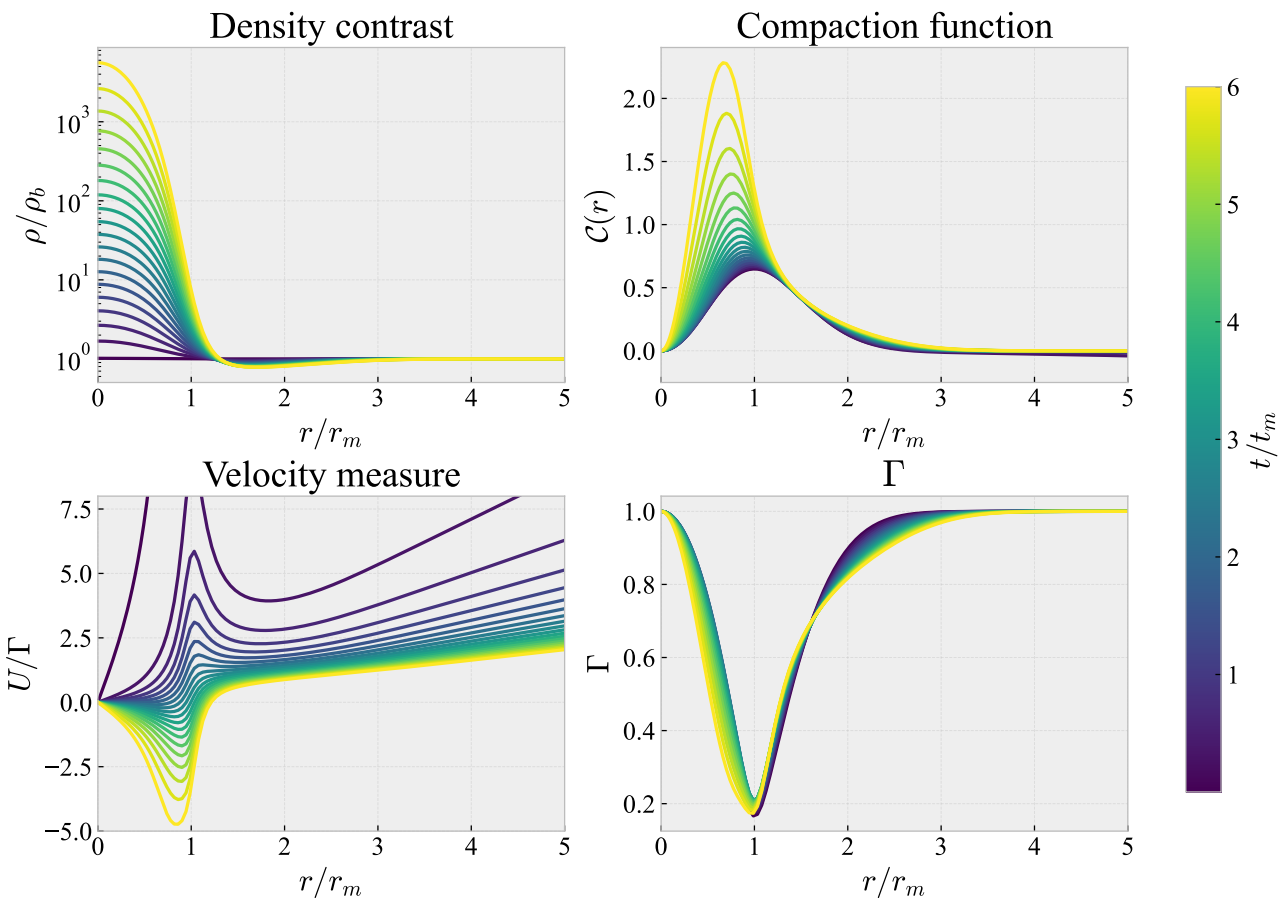

Figure 4.8: All profiles for extreme supercritical ($\bar{\delta}_m = 0.65$) case. Density (upper left), compaction function (upper right), velocity (lower left) and Γ (lower right).

4.1.3 Interpreting Rate of Collapse or Dispersion

Although we can assess the rate of dispersion/collapse at discrete time intervals by observing and tracking the peaks and the distance between them, fine structure and information may be hidden at this scale. An alternative approach to analysing the temporal behaviour of the system is through sonification, where numerical data is converted into sound signals. Despite being a relatively new method to study and present data (Kennel & Pizzera, 2016), the broader principle of employing multiple senses to interpret signals has long been present within scientific practice. The human ear is capable of recognising subtle changes and thus is ideal for studying such fine behaviour related to this system. Various software can map the data to notes by any instrument or octave. It is important to maintain an intuitive logic for the map, so we have selected a collapse to be mapped to increasing frequency and a dispersion to be mapped with decreasing frequency[2]. The attached sound and video files[3] contain the mapped notes to the increasing or decreasing peak as time evolves. In particular, we have selected the compaction function, as we are interested in qualitatively diagnosing on the rate of

[2]The main software we have used for producing the sound file associated with this project is https://sonification.highcharts.com

[3]Sonification files with animation can be found here https://github.com/meganhmartinez/PBHCapstone/tree/main/Sonification

collapse or dispersion of the perturbation. In order to notice this behaviour, we map the peak maxima to notes reflecting the relative change between amplitudes. In particular, for the supercritical case, we notice how the acceleration is not constant and that it must be a function of both time and another variable (mass or density). In particular, we notice the larger the peak is, the faster the subsequent peak will rise. We are studying discrete, but equidistant intervals of time within this model, but they can be as many or as few as we wish. On the contrary, for the subcritical case, the opposite behaviour can be heard. A rapid deceleration. Initial notes achieve a lower frequency relative to the previous note, while towards later times, they are more similar, solidifying our conclusion that a stable point is being reached. Therefore regardless of whether we choose increasing or decreasing frequency for either case, one should still be able to identify which regime we are in by the relative difference from one note to the next. Concisely, we found that the relative difference between successive peaks were larger in the beginning of the process for a dispersion and at the end of the process for a collapse. This initiated our search for quantifying this behaviour, and led to an attempt at fitting an exponential curve to the peaks of the compaction function at the same time intervals. This would potentially mean finding a rate of collapse which depends on the perturbation scale applied. Focusing on the collapse and following the logic of mapping the compaction function peaks, the maxima were plotted against time to observe an exponentially growing peak position as expected. The objective was to find a general fit to describe the collapsing compaction function for any suitable supercritical value. To initialise the study, we consider $\bar{\delta}_m = 0.5, 0.52, 0.54$ and we study the compaction function's peaks with time. After determining the best fit approach to use, we implement a polynomial fit to a general exponential curve in order to maintain dependence on δ. Thus we are not simply fitting a general $A + Be^{Ct}$, we are fitting a similar curve, where the fit coefficients are polynomials in δ. We obtain a long expression for $C_{\text{max peaks}}(t, \bar{\delta}_m)$ approximated by,

$$\begin{aligned} C_{\text{max peaks}}(t, \bar{\delta}_m) = &\left(-12.83875\,\bar{\delta}_m^2 + 15.184425\,\bar{\delta}_m - 3.890798\right) \\ &+ \left(-0.0675\,\bar{\delta}_m^2 + 0.1056\,\bar{\delta}_m - 0.034724\right) \\ &\times \exp\left(\left(-118.81125\,\bar{\delta}_m^2 + 131.244775\,\bar{\delta}_m - 35.694677\right) t\right). \end{aligned}$$

Comparing the plots generated from the peaks at particular times to the continuous general peak function, we find Figure 4.9 presents the general function aligning closely at small and large times relative to each collapse. The general function shows the behaviour for any supercritical value, but this figure serves as a reference for how the collapse differs for small changes in $\bar{\delta}_m$. As the critical threshold is approached, the fit is drawn to a horizontal line, as expected. By using sonification as a means of interpreting the results, it gave rise to new questions and results surrounding the supercritical behaviour in collapsing PBHs. Evidently, the collapse is increasingly sudden and rapid the larger $\bar{\delta}_m$ is. While similar studies can be mirrored for subcritical behaviour, we have focused on the collapse.

Of course, we are not limited to studying the maximum peak, or the compaction function for that matter. One could study the peak width at a particular value for a certain profile, one could map the radial position to another instrument, one could study the slope of the velocity or the distance between the maxima and minima when a split is seen within the fluid. Doing so allows us to isolate particular aspects within the behaviour and identify patterns or unique behaviour. In principle, if the

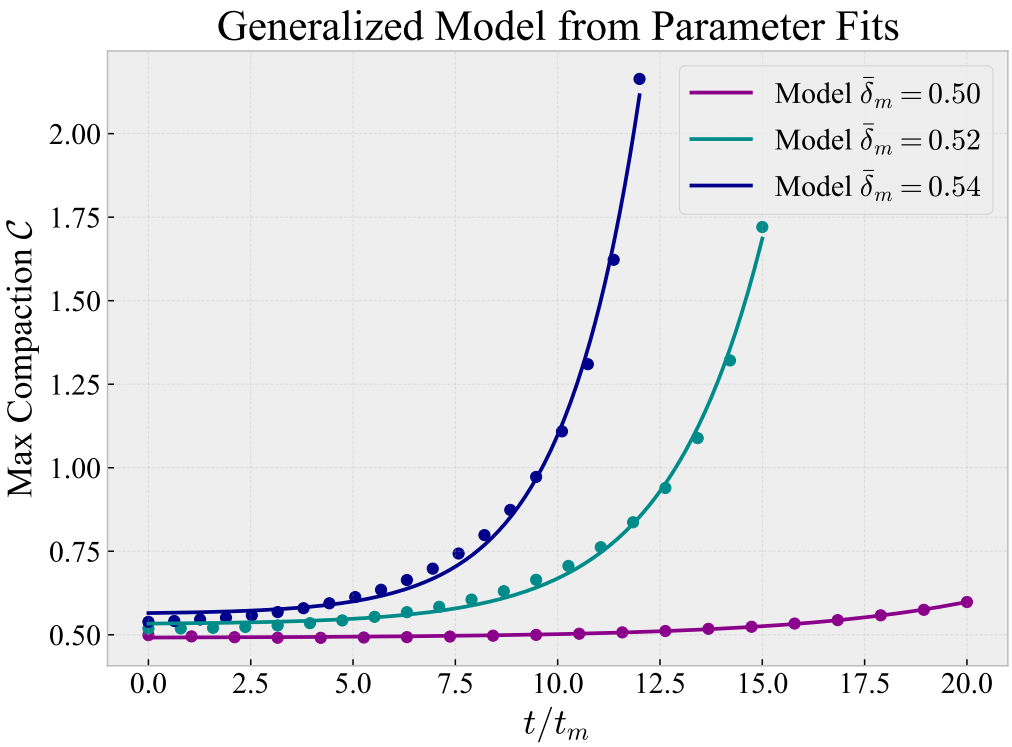

Figure 4.9: Plot with true peak values as points with general fit function of the same colour showing strong correlation with real data.

sonification map from separate measurements is overlayed, they should be describing the same process (a potential avenue for future work). While we focus specifically on the rate of collapse or dispersion, and this is a relatively simple model whose behaviour is well understood graphically, this tool renders itself useful to complement studies of such a system.

4.1.4 Changes and Efficiency

While the numerical Python code from Escrivà (2020) was publicly accessible, some changes were made for preference and convenience in order to create the plots. A major change which was made to the structure of the code was initialising a class. Object-oriented programming is convenient, in particular when there is a large number of variables and functions. Appendix D presents an excerpt for the class in the initial stages of the simulation. Note that the majority of the labelling of variables is consistent with the source code, which itself is more consistent with Polnarev & Musco (2007). The change to object-oriented programming also involved changing the structure of some functions, as their variables were now initial class variables. For example, the function for defining ϵ became simpler in our version with this change, and all initial class variables became self-contained. Some functions were combined into a larger parent function for readability and understanding structuring and order of the code. This also involved relabelling some functions and variables for readability. Some changes to efficiency of the code include taking a lower number of maximum Chebyshev polynomials (we have taken 200 instead of 400, while maintaining comparable graphs), and also changing the time tolerance from 10^{-5} to 10^{-3}. For overall structure and dynamics, having additional precision in the numerical values is not

necessary. Thus we have exchanged numerical precision for computational time efficiency.

4.2 Discussion

The numerical process above illustrates the dynamic behaviour of the particular variables with time following the perturbation, however uncertainty surrounding the numerical critical threshold value should be considered when interpreting the resulting evolution of the system. The critical threshold is sensitive to numerical resolution, as steep pressure gradients develop near the critical solution. Thus we are highly limited in terms of simulation abilities to interpret the collapse itself, along with any subsequent behaviour and evolution.

In comparison to existing alternative approaches to a similar problem, this method lends itself to cosmological perturbation theory and works naturally with the FRW background. Working in the co-moving space-like surfaces (or space-slicing) is convenient for the conditions we are studying, however limitations arise near the horizon formation. The alternative is called null-slicing because of its approach with evolving along null-hypersurfaces. This method operates neatly following radiation and the horizon formation instead. While both techniques have their limitations, they also have their strengths, and for the tasks we wish to accomplish here, the spectral approach is useful for the conditions and behaviour near criticality (Hernandez & Misner, 1966).

4.2.1 Error and Uncertainty

The primary source of uncertainty arises from the numerical determination of the critical threshold itself. As the perturbation approaches the critical regime, the system evolves toward a near self-similar solution, which is characterised by increasingly steep gradients in the variables. In particular, the ratio of spatial derivatives appearing in the Misner-Sharp equations, such as the term involving ρ'/R', becomes large as the collapse progresses. This behaviour places practical limits on the achievable numerical resolution, leading to instabilities when attempting to refine the threshold beyond a certain tolerance, bounded by 10^-5. Lower values were established by Escrivà (2020) to prove problematic in a numerical setting.

In addition, the bisection procedure used to determine the critical threshold introduces an inherent discretisation in the parameter space of initial perturbations. While the method is robust for locating the transition between dispersion and collapse, the final value of the threshold depends on the chosen tolerance and the numerical resolution of the evolution. As a result, the reported value of the critical threshold should be interpreted as an approximation within the achievable numerical precision rather than an exact value. As we achieve a critical threshold which aligns closely with literature, we can also conclude that the dominant source of error arises from limited resolution, as opposed to an inconsistent technique.

Moreover, we understand that the behaviour associated with the collapse while studying the super-critical case is further limited as the collapse happens. In particular, we encounter divisions by zero as $r = 0$ is approached. Therefore, we are restricted to studying the initial behaviour which indicates collapse.

Chapter 5

Conclusions and Future Work

In essence, we have verified the results presented by Escrivà (2020) through recreation and further analysis on the formation of primordial black holes within a cosmological setting by combining analytical methods from cosmological perturbation theory with numerical simulations based on the Misner-Sharp formalism. By assuming spherical symmetry and a radiation-dominated FRW background, we constructed a consistent framework to evolve super-horizon perturbations through horizon re-entry and determine their collapse. After establishing, with some precision, that $\delta_c \approx 0.49774 \pm 2 \times 10^{-5}$, we were capable of studying near critical behaviour, as well as extreme subcritical and supercritical cases. This value aligns both with Escrivà (2020) and existing literature under the same Gaussian curvature (Musco, 2019).

Furthermore, verification of the critical threshold has added credibility to the simulations which followed. We have obtained deeper insight to the dynamics and behaviour of our system resulting from the initial perturbation. We can conclude that the subcritical case disperses the perturbation. Ultimately, the profiles do return to the FRW background. When studied at the critical threshold, we do observe critical behaviour, in particular with respect to the compaction function. The critical case exhibits the boundary behaviour separating the dispersion and collapse. In contrast, the supercritical case allows us to study the rapid behaviour resulting in a collapse into a PBH. To summarise, these results give rise to an analytical level of understanding derived from the numerical basis. In addition, we have added a complementary technique for this analysis through the sonification, allowing for furthering our knowledge of the finer structure within the dispersion or collapse. This led to an original finding around the compaction function's peaks with time during a collapse, resulting in a fitted function to simulate this behaviour within the supercritical regime.

Eventually, one anticipates that numerical research contributing to determining the initial conditions resulting in PBHs to be observed and verified with alternative approaches. As PBHs are a candidate for dark matter, we expect that the observation of a PBH to be relevant both in solidifying the theories of the early Universe, while simultaneously providing answers on dark matter and star, black hole and galaxy formation. This particular project has the scope to extend our knowledge of such a collapse or dispersion by providing the tools for fine tuning the parameters we expect to find in the early Universe. Applying techniques such as sonification to complement our visual comprehension will add new results and aid in validating existing results and analysis.

Acknowledgements

MHM acknowledges the academic support of Professor Paschalis Karageorgis with the capstone research associated with the School of Mathematics, Trinity College Dublin. The authors of this project utilized the following software: NUMPY (Harris et al., 2020) and MATPLOTLIB (Hunter, 2007). We acknowledge the use of code provided by https://github.com/albert-escriva/SPriBHoS.

Appendix A

Differentiation Matrix General Form

This appendix derives the explicit form of the Chebyshev differentiation matrix $\mathbf{D}$ used in the numerical implementation, whose entries are evaluated at the collocation points and applied to compute spatial derivatives via $f' = Df$. Considering we are interested in an operation that works in the following way, namely a map from $\mathbb{P}_n \to \mathbb{P}_n$, which we shall write discretely from $\mathbb{R}^{n+1} \to \mathbb{R}^{n+1}$, where the vector components are $D : P(x_i) \mapsto P'(x_i)$. We achieve this with left multiplication with matrix $\mathbf{D}$, the differentiation matrix, in the following way,

$$\begin{bmatrix} P'(x_0) \\ \vdots \\ P'(x_n) \end{bmatrix} = \begin{bmatrix} D_{00} & \cdots & D_{0n} \\ \vdots & \ddots & \vdots \\ D_{n0} & \cdots & D_{nn} \end{bmatrix} \begin{bmatrix} P(x_0) \\ \vdots \\ P(x_n) \end{bmatrix}.$$

Thus, $P'(x_i) = \sum_j D_{ij} P(x_j)$ is true for any polynomial, and is exactly what has been applied to achieve Equation (3.47).

In order for this matrix to be translated to Python code following the association to the Chebyshev polynomials, it is simpler to manually evaluate the first derivative of Equation (3.45) at the collocation points. This implies,

$$D_{ab} = \frac{\bar{c}_a}{\bar{c}_b} \frac{(-1)^{a+b}}{(x_a - x_b)}, \quad (a \neq b), \quad a, b = 1, ..., N-1,$$

$$D_{aa} = -\frac{x_a}{2(1 - x_a^2)}, \quad a = 1, ..., N-1,$$

$$D_{00} = -D_{NN} = \frac{2N^2 + 1}{6}.$$

The diagonal terms can be compactly written as,

$$D_{aa} = -\sum_{b=0, b\neq a}^{N} D_{ab},$$

(Trefethen, 2000). Therefore, the code can now be implemented with a simple matrix. This information is compiled into the relevant code given by,

```
def chebymatrix(Ncheb,a,b):

  range_cheb = np.arange(0,Ncheb+1)
  x = np.cos(pi*range_cheb/Ncheb)
  t = (a+b)/2.-((a-b)/2.)*x
  carray = np.hstack([2, np.ones(Ncheb-1), 2])*(-1)**np.arange(0,Ncheb+1)
  X = np.tile(x,(Ncheb+1,1))
  dX = X.T - X
  Dp = (carray[:,np.newaxis]*(1.0/carray)[np.newaxis,:])/(dX+(np.identity(Ncheb+1)))
  Dp = Dp - np.diag(Dp.sum(axis=1))
  Dcheb =(2./(b-a))*Dp

  return Dcheb, t
```

Appendix B

Review δ_c

Result of running exact script code for implementing bisection method with the same initial conditions as Escrivà (2020), until a value of δ_c is approached to an error of 10^{-5}:

```
Welcome to the primordial black hole simulator,
The simulation is done with the following parameters:
N_cheb: 400
dt0: 0.001
Nh: 90
rm_N 10.0
error allowed in bisection 1e-05
The simulation is in process
We try the value of delta: 0.5333333333333333
Yes BH formation, we proceed with the next bisection iteration
We try the value of delta: 0.4666666666666667
No BH formation, we proceed with the next bisection iteration
We try the value of delta: 0.5
Yes BH formation, we proceed with the next bisection iteration
We try the value of delta: 0.48333333333333334
No BH formation, we proceed with the next bisection iteration
We try the value of delta: 0.4916666666666667
No BH formation, we proceed with the next bisection iteration
We try the value of delta: 0.49583333333333335
No BH formation, we proceed with the next bisection iteration
We try the value of delta: 0.4979166666666667
Yes BH formation, we proceed with the next bisection iteration
We try the value of delta: 0.496875
No BH formation, we proceed with the next bisection iteration
We try the value of delta: 0.49739583333333337
No BH formation, we proceed with the next bisection iteration
We try the value of delta: 0.49765625
```

```
No BH formation, we proceed with the next bisection iteration
We try the value of delta: 0.49778645833333335
Yes BH formation, we proceed with the next bisection iteration
We try the value of delta: 0.4977213541666667
No BH formation, we proceed with the next bisection iteration
We try the value of delta: 0.49775390625000004
Yes BH formation, we proceed with the next bisection iteration
The value of the threshold and its resolution is given by:
0.4977376302083334 1.6276041666651864e-05
Simulation done successfully. The time of the computation was:
--- 788.2134480476379 seconds ---
```

Appendix C

Initial Values for Simulation

Parameter	**Value**	**Description**
t_0	1	Initial reference time (dimensionless units)
$s(t_0) = s_0$	1	Initial scale factor
H_0	1/2	Initial Hubble parameter
$R_H(t_0)$	2	Initial cosmological horizon radius
r_m	$10R_H(t_0)$	Characteristic perturbation scale
t_m	$10^2 t_0$	Time when perturbation scale equals horizon
ω	1/3	Equation-of-state parameter (radiation dominated Universe)
α	1/2	Scale factor exponent $s(t) \propto t^\alpha$
N_{cheb}	400 or 200	Number of Chebyshev collocation points for bisection or plot simulations
dt_0	10^{-3}	Initial time-step for Runge–Kutta evolution
q	1	Gaussian curvature profile parameter

Table C.1: Numerical parameters and initial conditions used in the simulations. These values follow the setup of Escrivà (2020) to allow direct comparison with the published results.

Appendix D

Class Excerpt for Initialising Simulation

```
class PBHSimulator:
    def __init__(self, p: SimParams):
        self.p = p
        self.alpha = 2.0 / (3.0 * (1.0 + p.w))
        self.H0 = self.alpha / p.t0
        self.RH0 = 1.0 / self.H0
        self.eb0 = (3.0 / (8.0 * pi)) * (self.H0 ** 2)

        self.rmin = 0.0
        self.rmax = p.Nh * self.RH0

        # Chebyshev grid + D
        self.D, self.r = chebymatrix(p.Ncheb, self.rmin, self.rmax)
        self.ones = np.ones(p.Ncheb + 1, dtype=np.float64)

        #Previously defined within epsilon horizon crossing
        self.rm = p.rm_N * self.RH0
        self.tm = p.t0 * (p.rm_N ** 2)
```

Appendix E

Full Code

The following is all of the code used for graphing and studying PBHs:

```
import math
import numpy as np
import matplotlib.pyplot as plt
from dataclasses import dataclass
from typing import Dict, List, Tuple, Optional
import seaborn as sns
cmap = sns.color_palette("viridis", as_cmap=True)
pi = math.pi

def chebymatrix(Ncheb,a,b):

  range_cheb = np.arange(0,Ncheb+1)
  x = np.cos(pi*range_cheb/Ncheb)
  t = (a+b)/2.-((a-b)/2.)*x
  carray = np.hstack([2, np.ones(Ncheb-1), 2])*(-1)**np.arange(0,Ncheb+1)
  X = np.tile(x,(Ncheb+1,1))
  dX = X.T - X
  Dp = (carray[:,np.newaxis]*(1.0/carray)[np.newaxis,:])/(dX+(np.identity(Ncheb+1)))
  Dp = Dp - np.diag(Dp.sum(axis=1))
  Dcheb =(2./(b-a))*Dp

  return Dcheb, t

def curvature_profile(xf,rmf):
  K = np.exp(-(xf/rmf)**2)
  return K

def derivative_curvature_profile(xf,rmf):
  Kd = -2.*(xf/rmf**2)*np.exp(-(xf/rmf)**2)
  return Kd


def A_versus_threshold_original_style(rm, w, delta_bar_m) :

    fw = 3.0 * (1.0 + w) / (5.0 + 3.0 * w)
```

```
    Krm = math.exp(-1.0)  # Gaussian Kbar(rm)=e^{-1}
    return delta_bar_m / (fw * Krm * (rm ** 2))


def A_versus_threshold_general(rm, w, delta_bar_m) :
    fw = 3.0 * (1.0 + w) / (5.0 + 3.0 * w)
    Kbar_rm = math.exp(-(1.0 ) * (1.0 ** (2.0 )))
    return delta_bar_m / (fw * Kbar_rm * (rm ** 2))


@dataclass
class SimParams:
    w: float = 1.0 / 3.0
    t0: float = 1.0
    a0: float = 1.0
    Nh: int = 90
    rm_N: float = 10.0
    dt0: float = 1e-3
    Ncheb: int = 200
    t_final: float = 80000
    use_general_q: bool = False


# Same initial conditions, just defined as a class instead
class PBHSimulator:
    def __init__(self, p: SimParams):
        self.p = p
        self.alpha = 2.0 / (3.0 * (1.0 + p.w))
        self.H0 = self.alpha / p.t0
        self.RH0 = 1.0 / self.H0
        self.eb0 = (3.0 / (8.0 * pi)) * (self.H0 ** 2)

        self.rmin = 0.0
        self.rmax = p.Nh * self.RH0

        # Chebyshev grid + D
        self.D, self.r = chebymatrix(p.Ncheb, self.rmin, self.rmax)
        self.ones = np.ones(p.Ncheb + 1, dtype=np.float64)

        #Previously defined within epsilon horizon crossing
        self.rm = p.rm_N * self.RH0
        self.tm = p.t0 * (p.rm_N ** 2)


    def energy_FRW(self, t) :
        return self.ones * self.eb0 * (self.p.t0 / t) ** 2

    def solution_FRW(self, t) :
        a = self.p.a0 * (t / self.p.t0) ** self.alpha   #scale factor
        H = self.alpha / t                              #Hubble Factor
        e = self.energy_FRW(t)                          #Energy density of the
```

```
                                                    background
    R = a * self.r                                        #Areal Radius
    U = R * H                                             #Euler Velocity
    M = (4.0 * pi / 3.0) * e * R ** 3                     #Mass of Background
    A = 1.0 * self.ones                                   #Lapse Function
    G = 1.0 * self.ones                                   #Gamma Function
    return e, R, U, M, A, G

def initial_perturbation_magnitudes(self, K, Kp) :
    w = self.p.w
    dek = (3.0 * (1.0 + w) / (5.0 + 3.0 * w)) * (K + (self.r / 3.0) * Kp) * (self
                                                    .rm ** 2)
    dUk = -(1.0 / (5.0 + 3.0 * w)) * K * (self.rm ** 2)
    dAk = -(w / (1.0 + w)) * dek
    dMk = -3.0 * (1.0 + w) * dUk
    dRk = -dek * w / ((1.0 + 3.0 * w) * (1.0 + w)) + dUk / (1.0 + 3.0 * w)
    return dek, dUk, dAk, dMk, dRk

def epsilon_horizon_crossing(self):
    eps = 1.0 / (self.p.a0 * self.H0 * self.rm)
    tH = self.tm
    return eps, tH

#called x_Ikk in source code
def initial_conditions(self, eps, dek, dRk, dUk, dAk, dMk):
    eI = self.eb0 * (self.ones + dek * (eps ** 2))
    aI = self.p.a0
    RI = aI * self.r * (self.ones + dRk * (eps ** 2))
    UI = self.H0 * RI * (self.ones + dUk * (eps ** 2))
    AI = self.ones + dAk * (eps ** 2)
    MI = (4.0 * pi / 3.0) * self.eb0 * (self.ones + dMk * (eps ** 2)) * RI ** 3
    frac = MI[:-1] / RI[:-1]
    frac = np.insert(frac, len(frac), 0.0)
    GI = np.sqrt(self.ones + UI ** 2 - 2.0 * frac)
    return eI, RI, UI, AI, MI, GI

def derivs(self, e, U, R) :
    de = np.dot(self.D,e)
    dU = np.dot(self.D,U)
    dR = np.dot(self.D,R)
    return de, dU, dR

def system_static(self, e, M, R, U, eFRW):
    w = self.p.w
    A = (eFRW / e) ** (w / (w + 1.0))
    frac = M[:-1] / R[:-1]
    frac = np.insert(frac, len(frac), 0.0)

    arg = 1.0 + U**2 - 2.0 * frac

    # numerical guard: avoid tiny negative values from roundoff
```

```
        arg = np.maximum(arg, 0.0)

        G = np.sqrt(arg)
        return A, G

    #compiling excision functions from source code
    def RK_excision(self, U, R, M, A, G, e, de, dR, dU) :
        w = self.p.w
        frac = M[:-1] / (R[:-1] ** 2)
        frac = np.insert(frac, len(frac), 0.0)
        Ut = -A * (4.0 * pi * R * w * e + frac + (w / (1.0 + w)) * (de * G ** 2) / (e
                                                                    * dR))
        Rt = U * A
        Mt = -A * 4.0 * pi * w * e * U * (R ** 2)

        derU_R = dU / dR
        ratioUR = U[:-1] / R[:-1]
        ratioUR = np.insert(ratioUR, len(ratioUR), derU_R[-1])
        et = -A * e * (1.0 + w) * (2.0 * ratioUR + dU / dR)
        return Ut, Rt, Mt, et

    #compiling computation executing the solving, where stages represent different
                                                        functions in source code
    def computation_excision(self, t, dt, U, R, M, A, G, e):
        # stage 1
        de1, dU1, dR1 = self.derivs(e, U, R)
        de1[0] = 0.0
        de1[-1] = 0.0
        k1U, k1R, k1M, k1e = self.RK_excision(U, R, M, A, G, e, de1, dR1, dU1)

        # stage 2
        e2 = e + 0.5 * dt * k1e
        U2 = U + 0.5 * dt * k1U
        R2 = R + 0.5 * dt * k1R
        M2 = M + 0.5 * dt * k1M
        U2[-1] = R2[-1] = M2[-1] = 0.0

        eFRW2 = self.energy_FRW(t + 0.5 * dt - dt)
        A2, G2 = self.system_static(e2, M2, R2, U2, eFRW2)
        de2, dU2, dR2 = self.derivs(e2, U2, R2)
        de2[0] = 0.0
        de2[-1] = 0.0
        k2U, k2R, k2M, k2e = self.RK_excision(U2, R2, M2, A2, G2, e2, de2, dR2, dU2)

        # stage 3
        e3 = e + 0.5 * dt * k2e
        U3 = U + 0.5 * dt * k2U
        R3 = R + 0.5 * dt * k2R
        M3 = M + 0.5 * dt * k2M
        U3[-1] = R3[-1] = M3[-1] = 0.0
```

```
        eFRW3 = self.energy_FRW(t + 0.5 * dt - dt)
        A3, G3 = self.system_static(e3, M3, R3, U3, eFRW3)
        de3, dU3, dR3 = self.derivs(e3, U3, R3)
        de3[0] = 0.0
        de3[-1] = 0.0
        k3U, k3R, k3M, k3e = self.RK_excision(U3, R3, M3, A3, G3, e3, de3, dR3, dU3)

        # stage 4
        e4 = e + dt * k3e
        U4 = U + dt * k3U
        R4 = R + dt * k3R
        M4 = M + dt * k3M
        U4[-1] = R4[-1] = M4[-1] = 0.0

        eFRW4 = self.energy_FRW(t + dt - dt)
        A4, G4 = self.system_static(e4, M4, R4, U4, eFRW4)
        de4, dU4, dR4 = self.derivs(e4, U4, R4)
        de4[0] = 0.0
        de4[-1] = 0.0
        k4U, k4R, k4M, k4e = self.RK_excision(U4, R4, M4, A4, G4, e4, de4, dR4, dU4)

        # combine
        Unew = U + (dt / 6.0) * (k1U + 2 * k2U + 2 * k3U + k4U)
        Rnew = R + (dt / 6.0) * (k1R + 2 * k2R + 2 * k3R + k4R)
        Mnew = M + (dt / 6.0) * (k1M + 2 * k2M + 2 * k3M + k4M)
        enew = e + (dt / 6.0) * (k1e + 2 * k2e + 2 * k3e + k4e)

        Unew[-1] = 0.0
        Rnew[-1] = 0.0
        Mnew[-1] = 0.0

        eFRW = self.energy_FRW(t + dt)
        Anew, Gnew = self.system_static(enew, Mnew, Rnew, Unew, eFRW)
        return Unew, Rnew, Mnew, enew, Anew, Gnew, eFRW

    #Redefined in a simpler way
    def compaction(self, M, R, eFRW ) :
        C = 2.0 * (M[:-1] - (4.0 / 3.0) * pi * eFRW[:-1] * (R[:-1] ** 3)) / R[:-1]

        C = np.insert(C, -1, 0.0)
        return C

    def build_initial_state(self, delta_bar_m) :
        eps, _ = self.epsilon_horizon_crossing()


        Kbar = curvature_profile(self.r, self.rm)
        Kbar_d = derivative_curvature_profile(self.r, self.rm)
        Aamp = A_versus_threshold_general(self.rm, self.p.w, delta_bar_m)

        K = Aamp * Kbar
```

```
        Kd = Aamp * Kbar_d

        dek, dUk, dAk, dMk, dRk = self.initial_perturbation_magnitudes(K, Kd)
        eI, RI, UI, AI, MI, GI = self.initial_conditions(eps, dek, dRk, dUk, dAk, dMk
                                                        )
        return eI, RI, UI, AI, MI, GI
    #this involves original definitions for plotting purposes
    def evolve_snapshots(self, delta_bar_m, snapshot_times_over_tm,t_stop_over_tm:
                                                Optional[float] = None):

        e, R, U, A, M, G = self.build_initial_state(delta_bar_m)
        t = self.p.t0

        snapshot_times_over_tm = sorted(snapshot_times_over_tm)
        targets = [tau * self.tm for tau in snapshot_times_over_tm]
        target_idx = 0

        out: Dict[float, Dict[str, np.ndarray]] = {}
        eps, tH = self.epsilon_horizon_crossing()

        # stop time
        if t_stop_over_tm is None:
            t_stop = targets[-1]
        else:
            t_stop = t_stop_over_tm * self.tm

        # main loop
        while t < t_stop and target_idx < len(targets):
            dt = self.p.dt0 * math.sqrt(t)
            t_next = t + dt

            U, R, M, e, A, G, eFRW = self.computation_excision(t, dt, U, R, M, A, G,
                                                            e)
            t = t_next

            while target_idx < len(targets) and t >= targets[target_idx]:

                tau = snapshot_times_over_tm[target_idx]

                # quantities used in Figs 4âĂŞ6
                # rho/rho_b
                rho_ratio = e / eFRW

                # compaction C
                C = self.compaction(M, R, eFRW)

                Gamma = G.copy()

                # U/Gamma
                with np.errstate(divide="ignore", invalid="ignore"):
                    U_over_Gamma = np.where(Gamma != 0.0, U / Gamma, 0.0)
```

```
                # store
                r_over_rm = self.r / self.rm

                out[tau] = {
                    "r": self.r.copy(),
                    "rm": np.array([self.rm], dtype=float),
                    "r_over_rm": r_over_rm.copy(),

                    "R": R.copy(),
                    "U": U.copy(),
                    "M": M.copy(),
                    "A": A.copy(),
                    "Gamma": Gamma,
                    "rho": e.copy(),
                    "rho_FRW": eFRW.copy(),
                    "rho_ratio": rho_ratio,
                    "C": C,
                    "U_over_Gamma": U_over_Gamma,
                    "t": np.array([t], dtype=float),
                    "t_over_tm": np.array([t / self.tm], dtype=float),
                    }


                target_idx += 1

        return out


def _paper_axes_style(ax: plt.Axes, xlabel, ylabel, title):
    ax.set_xlabel(xlabel)
    ax.set_ylabel(ylabel)
    if title:
        ax.set_title(title)
    ax.grid(True, alpha=0.3)

def plot_profiles_single(
    snapshots,
    *,
    delta_bar_m,
    label,
    add_colorbar=False,
    pdf=None,
):

    taus = sorted(snapshots.keys())
    norm = plt.Normalize(vmin=min(taus), vmax=max(taus))

    profiles = [
        ("rho_ratio", plt.semilogy, r"$\rho/\rho_b$", "Density contrast"),
        ("C", plt.plot, r"$\mathcal{C}(r)$", "Compaction function"),
```

```
        ("U_over_Gamma", plt.plot, r"$U/\Gamma$", "Velocity measure"),
        ("Gamma", plt.plot, r"$\Gamma$", r"$\Gamma$"),
    ]

    for key, plot_func, ylabel, title in profiles:

        # make supercritical plots wider to compensate for colorbar
        if add_colorbar:
            fig, ax = plt.subplots(figsize=(7.5,5))
        else:
            fig, ax = plt.subplots(figsize=(6,5))

        for tau in taus:

            d = snapshots[tau]

            x = d["r_over_rm"]
            order = np.argsort(x)
            x = x[order]

            y = d[key][order]

            color = cmap(norm(tau))
            plot_func(x, y, color=color)

        ax.set_xlim(0,5)

        if key == "U_over_Gamma":
            ax.set_ylim(-2,5)

        _paper_axes_style(ax, r"$r/r_m$", ylabel, None)

        fig.suptitle(f"{title} - {label} ($\\bar\\delta_m$={delta_bar_m})")

        # colorbar only on supercritical

        sm = plt.cm.ScalarMappable(cmap=cmap, norm=norm)

        sm.set_array([])
        cbar = fig.colorbar(sm, ax=ax)
        cbar.set_label(r"$t/t_m$")

        fig.tight_layout()

        if pdf:
            pdf.savefig(fig)

        plt.close(fig)
def main():

    p = SimParams(
```

```
        w=1.0 / 3.0,
        t0=1.0,
        a0=1.0,
        Nh=90,
        rm_N=10.0,
        dt0=1e-3,
        Ncheb=200,
        t_final=8.0e4,
        use_general_q=False,
    )

    sim = PBHSimulator(p)

    delta_vals = [0.49, 0.49775, 0.51]
    labels = ["Subcritical", "Critical", "Supercritical"]

    snapshot_taus = [0.01,3,9,13,17,19,24,27,32,38,40,44]

    with PdfPages("/Users/Megan/Downloads/SPriBHoS/plots2.pdf") as pdf:

        for delta_bar_m, label in zip(delta_vals, labels):

            snaps = sim.evolve_snapshots(
                delta_bar_m=delta_bar_m,
                snapshot_times_over_tm=snapshot_taus
            )

            plot_profiles_single(
                snaps,
                delta_bar_m=delta_bar_m,
                label=label,
                add_colorbar=(label == "Supercritical"),
                pdf=pdf
            )

if __name__ == "__main__":
    main()
```

Code for finding the general function to describe the collapse as a means of studying the compaction function is found here:

```
#test delta data
delta_data = np.array([0.50, 0.52, 0.54])

C0_data = np.array([0.491727, 0.533505, 0.565012])
B_data  = np.array([0.001201, 0.001936, 0.002617])
lam_data = np.array([0.224898, 0.426044, 0.532141])

#fit parameter in delta, exact for 3 points, but gives a formula for any delta in the
                                                    range,
# AI was used to determine the best fitting method and to implement it, but the
```

```
                                                actual fitting was done by hand and
                                                verified graphically
coef_C0  = np.polyfit(delta_data, C0_data, 2)
coef_B   = np.polyfit(delta_data, B_data, 2)
coef_lam = np.polyfit(delta_data, lam_data, 2)

def C0_of_delta(delta):
    return np.polyval(coef_C0, delta)

def B_of_delta(delta):
    return np.polyval(coef_B, delta)

def lam_of_delta(delta):
    return np.polyval(coef_lam, delta)

#to generalise the formula for any delta, we can use the fitted parameters to predict
                                                 C0, B and lambda, and then use those in
                                                the original formula
def C_model(t, delta):
    C0 = C0_of_delta(delta)
    B = B_of_delta(delta)
    lam = lam_of_delta(delta)
    return C0 + B * (np.exp(lam * t))

#print the fitted coefficients for reference
print("C0(delta) coefficients:", coef_C0)
print("B(delta) coefficients:", coef_B)
print("lambda(delta) coefficients:", coef_lam)

#an example of using the model to predict the compaction curve for a new delta value
delta_new = 0.4977
t = np.linspace(0, 6, 300)
C_pred = C_model(t, delta_new)

plt.figure(figsize=(8, 6))
plt.plot(t, C_pred, color='black', lw=2, label=rf"Predicted $\bar{{\delta}}_m={
                                                delta_new:.2f}$")
plt.xlabel(r"$t/t_m$")
plt.ylabel(r"Max Compaction $\mathcal{C}$")
plt.title("Generalized Predicted Collapse Curve")
plt.legend()
plt.grid(True, alpha=0.3)
plt.tight_layout()
plt.show()
```

All plots are generated using excerpts of the above code. All code and matching graphs are linked above in the GitHub repository.

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